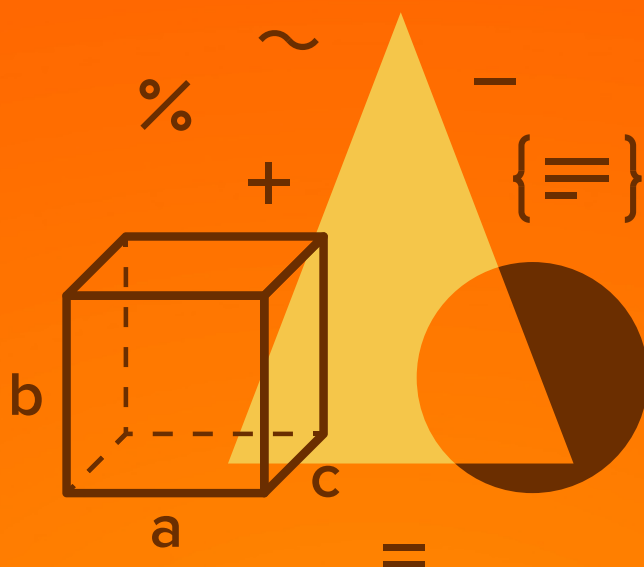




MATHEMATICS

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

Application of Derivative

English Medium

EXERCISE (O-1)

Straight Objective Type

1. If the surface area of a sphere of radius r is increasing uniformly at the rate $8\text{cm}^2/\text{s}$, then the rate of change of its volume is :

(A) proportional to r^2 (B) constant (C) proportional to r (D) proportional to $\sqrt{r}$

TN0001

2. A Spherical balloon is being inflated at the rate of 35 cc/min . The rate of increase in the surface area (in cm^2/min .) of the balloon when its diameter is 14 cm , is :

(A) $\sqrt{10}$ (B) $10\sqrt{10}$ (C) 100 (D) 10

TN0002

3. Let S be a square with sides of length x . If we approximate the change in size of the area of S by $h \cdot \frac{dA}{dx} \Big|_{x=x_0}$,

when the sides are changed from x_0 to $x_0 + h$, then the absolute value of the error in our approximation, is

(A) h^2 (B) $2hx_0$ (C) x_0^2 (D) h

TN0003

4. The slope of the curve $y = \sin x + \cos^2 x$ is zero at the point, where-

(A) $x = \frac{\pi}{4}$ (B) $x = \frac{\pi}{2}$ (C) $x = \pi$ (D) No where

TN0004

5. The equation of tangent at the point (at^2, at^3) on the curve $ay^2 = x^3$ is-

(A) $3tx - 2y = at^3$ (B) $tx - 3y = at^3$ (C) $3tx + 2y = at^3$ (D) None of these

TN0005

6. The equation of normal to the curve $y = x^3 - 2x^2 + 4$ at the point $x = 2$ is-

(A) $x + 4y = 0$ (B) $4x - y = 0$ (C) $x + 4y = 18$ (D) $4x - y = 18$

TN0006

7. The slope of the normal to the curve $x = a(\theta - \sin\theta)$, $y = a(1 - \cos\theta)$ at point $\theta = \pi/2$ is -

(A) 0 (B) 1 (C) -1 (D) $1/\sqrt{2}$

TN0007

8. The coordinates of the points on the curve $x = a(\theta + \sin\theta)$, $y = a(1 - \cos\theta)$, where tangent is inclined an angle $\pi/4$ to the x -axis are -

(A) (a, a) (B) $\left(a\left(\frac{\pi}{2}-1\right), a\right)$ (C) $\left(a\left(\frac{\pi}{2}+1\right), a\right)$ (D) $\left(a, a\left(\frac{\pi}{2}+1\right)\right)$

TN0008

9. Consider the curve represented parametrically by the equation
 $x = t^3 - 4t^2 - 3t$ and $y = 2t^2 + 3t - 5$ where $t \in \mathbb{R}$.
 If H denotes the number of point on the curve where the tangent is horizontal and V the number of point where the tangent is vertical then
 (A) $H = 2$ and $V = 1$ (B) $H = 1$ and $V = 2$ (C) $H = 2$ and $V = 2$ (D) $H = 1$ and $V = 1$
TN0009
10. The line $x/a + y/b = 1$ touches the curve $y = be^{-x/a}$ at the point -
 (A) $(0, a)$ (B) $(0, 0)$ (C) $(0, b)$ (D) $(b, 0)$
TN0010
11. If the tangent to the curve $2y^3 = ax^2 + x^3$ at a point (a, a) cuts off intercepts p and q on the coordinates axes, where $p^2 + q^2 = 61$, then a equals -
 (A) 30 (B) -30 (C) 0 (D) ± 30
TN0011
12. The sum of the intercepts made by a tangent to the curve $\sqrt{x} + \sqrt{y} = 4$ at point $(4, 4)$ on coordinate axes is -
 (A) $4\sqrt{2}$ (B) $6\sqrt{3}$ (C) $8\sqrt{2}$ (D) $\sqrt{256}$
TN0012
13. The curve $x^2 - y^2 = 5$ and $\frac{x^2}{18} + \frac{y^2}{8} = 1$ cut each other at any common point at an angle -
 (A) $\pi/4$ (B) $\pi/3$ (C) $\pi/2$ (D) None of these
TN0013
14. The lines tangent to the curve $y^3 - x^2y + 5y - 2x = 0$ and $x^4 - x^3y^2 + 5x + 2y = 0$ at the origin intersect at an angle θ equal to -
 (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{2}$
TN0014
15. If the tangent at a point P, with parameter t, on the curve $x = 4t^2 + 3$, $y = 8t^3 - 1$, $t \in \mathbb{R}$, meets the curve again at a point Q, then the coordinates of Q are :
 (A) $(t^2 + 3, t^3 - 1)$ (B) $(t^2 + 3, -t^3 - 1)$
 (C) $(16t^2 + 3, -64t^3 - 1)$ (D) $(4t^2 + 3, -8t^3 - 1)$
TN0015
16. The angle of intersection between the curves $y^2 = 8x$ and $x^2 = 4y - 12$ at $(2, 4)$ is -
 (A) 90° (B) 60° (C) 45° (D) 0°
TN0016
17. The length of subtangent to the curve $x^2 + xy + y^2 = 7$ at the point $(1, -3)$ is -
 (A) 3 (B) 5 (C) 15 (D) $3/5$
TN0017

18. For a curve $\frac{(\text{length of normal})^2}{(\text{length of tangent})^2}$ is equal to -

- (A) (subnormal)/(subtangent) (B) (subtangent)/(subnormal)
(C) (subtangent $\times$ subnormal) (D) constant

TN0018

19. At any point of a curve (subtangent) $\times$ (subnormal) is equal to the square of the-

- (A) slope of the tangent at that point (B) slope of the normal at that point
(C) abscissa of that point (D) ordinate of that point

TN0019

20. Let S be a sphere with radius r. If we approximate the change of volume of S by $h \cdot A \Big|_{r=r_0} + \frac{h^2}{2} \frac{dA}{dr} \Big|_{r=r_0}$ where A is surface area, when radius is changed from r_0 to $(r_0 + h)$, then the absolute value of error in our approximation is

- (A) h^3 (B) $4\pi r_0^2 h^2$ (C) $4\pi r_0 h^2$ (D) $\frac{4\pi}{3} h^3$

TN0020

21. At the point P(a, a^n) on the graph of $y = x^n$ ($n \in \mathbb{N}$) in the first quadrant a normal is drawn. The normal intersects the y-axis at the point (0, b). If $\lim_{a \rightarrow 0} b = \frac{1}{2}$, then n equals

- (A) 1 (B) 3 (C) 2 (D) 4

TN0021

22. A circle with centre at (15, -3) is tangent to $y = \frac{x^2}{3}$ at a point in the first quadrant. The radius of the circle is equal to

- (A) $5\sqrt{6}$ (B) $8\sqrt{3}$ (C) $9\sqrt{2}$ (D) $6\sqrt{5}$

TN0022

23. A line L is perpendicular to the curve $y = \frac{x^2}{4} - 2$ at its point P and passes through (10, -1). The coordinates of the point P are

- (A) (2, -1) (B) (6, 7) (C) (0, -2) (D) (4, 2)

TN0023

24. The point(s) at each of which the tangents to the curve $y = x^3 - 3x^2 - 7x + 6$ cut off on the positive semi axis OX a line segment half that on the negative semi axis OY then the co-ordinates of the point(s) is/are given by :

- (A) (-1, 9) (B) (3, -15) (C) (1, -3) (D) none

TN0024

25. The equation of the tangent to the curve $y = x + \frac{4}{x^2}$, that is parallel to the x-axis, is :-

- (A) $y = 0$ (B) $y = 1$ (C) $y = 2$ (D) $y = 3$

TN0068

EXERCISE (O-2)

Multiple Correct Answer Type

1. If $\frac{x}{a} + \frac{y}{b} = 1$ is a tangent to the curve $x = Kt$, $y = \frac{K}{t}$, $K > 0$ then :
- (A) $a > 0$, $b > 0$ (B) $a > 0$, $b < 0$ (C) $a < 0$, $b > 0$ (D) $a < 0$, $b < 0$

TN0029

2. The co-ordinates of the point(s) on the graph of the function, $f(x) = \frac{x^3}{3} - \frac{5x^2}{2} + 7x - 4$ where the tangent drawn cut off intercepts from the co-ordinate axes which are equal in magnitude but opposite in sign, is
- (A) $(2, 8/3)$ (B) $(3, 7/2)$ (C) $(1, 5/6)$ (D) none

TN0030

3. Given that $g(x)$ is a non constant linear function defined on $\mathbb{R}$ -
- (A) $y = g(x)$ and $y = g^{-1}(x)$ are orthogonal (B) $y = g(x)$ and $y = g^{-1}(-x)$ are orthogonal
- (C) $y = g(-x)$ and $y = g^{-1}(x)$ are orthogonal (D) $y = g(-x)$ and $y = g^{-1}(-x)$ are orthogonal

TN0031

4. If the curves $y = 2(x - a)^2$ and $y = e^{2x}$ touches each other, then 'a' is less than-
- (A) -1 (B) 0 (C) 1 (D) 2

TN0032

5. For the curve $C : y = e^{2x} \cos x$, which of the following statement(s) is/are true ?
- (A) equation of the tangent where C crosses y -axis is $y = 3x + 1$
- (B) equation of the tangent where C crosses y -axis is $y = 2x + 1$
- (C) number of points in $\left(-\frac{\pi}{2}, \frac{3\pi}{2}\right)$ where tangent on the curve C is parallel to x -axis is 4.
- (D) number of points in $\left(-\frac{\pi}{2}, \frac{3\pi}{2}\right)$ where tangent on the curve C is parallel to x -axis is 2.

TN0033

6. If tangent to curve $2y^3 = ax^2 + x^3$ at point (a, a) cuts off intercepts α, β on co-ordinate axes, where $\alpha^2 + \beta^2 = 61$, then the value of 'a' is equal to
- (A) 20 (B) 25 (C) 30 (D) -30

TN0083

7. For the curve $x = t^2 + 3t - 8$, $y = 2t^2 - 2t - 5$, at point $(2, -1)$
- (A) length of subtangent is $7/6$ (B) slope of tangent = $6/7$
- (C) length of tangent = $\sqrt{(85)}/6$ (D) none of these

TN0084

8. Given the curve $\left(\frac{x^2 - y^2}{xy}\right)^3 = \frac{512}{27}$, then

- (A) the slope of normal drawn to the curve at $(3,1)$ is $\frac{1}{3}$
- (B) the value of $\left(2\frac{dy}{dx} - 5\frac{d^2y}{dx^2}\right)_{(3,1)} = \frac{2}{3}$
- (C) the slope of normal drawn to the curve at $(3,1)$ is -3 .
- (D) the equation of tangent at $(3,1)$ to given curve is $3y + x = 0$.

TN0085

9. If the line $y = 1 - 4x$ touches the curve $y = x^3 + ax^2 - bx + c$ at $(0,1)$ and also touches the curve $y = \frac{x^2}{2} - px + 3$ at (α, β) , then

- (A) $b + c = 5$ (B) sum of all possible value(s) of p is 8
- (C) sum of all possible value(s) of α is 0 (D) sum of all possible value(s) of β is 2

TN0086

10. The equation of normal to curve $\left(\frac{x}{a}\right)^n + \left(\frac{y}{b}\right)^n = 2$, where $n \in \mathbb{N}$, at the point with abscissa equal to 'a' can be -

- (A) $ax + by = a^2 - b^2$ (B) $ax + by = a^2 + b^2$
- (C) $ax - by = a^2 - b^2$ (D) $bx + ay = a^2 - b^2$

TN0087

11. If curves $f(x) = x^2 + ax + b$ and $g(x) = cx - x^2$ touches each other at point $(1, 0)$, then
- (A) $a = -3$ (B) $b = 2$ (C) $c = 1$ (D) $abc = -6$

TN0088

12. If tangent at point $P(1, -8)$ on curve $y = x^3 - 9x$ intersects again at point Q (other than P) on curve, then-
- (A) Equation of tangent at P is $6x + y + 2 = 0$
- (B) Equation of tangent at Q is $3x - y + 16 = 0$
- (C) Equation of tangent with least slope is $9x + y = 0$
- (D) Slope of tangent where curve intersects x-axis can be 3

TN0089

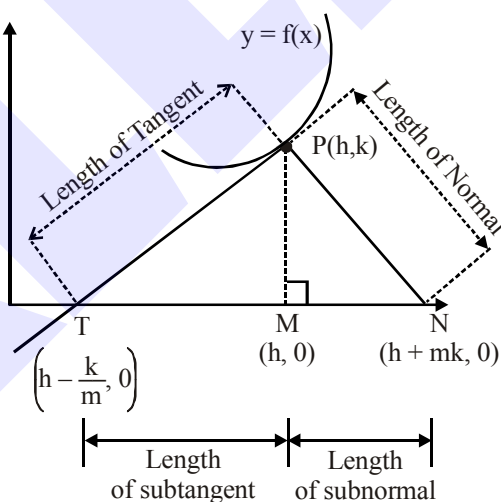
13. Tangents are drawn to the curve $y^2 - 2x^3 - 4y + 8 = 0$ from $(1, 2)$, then
 (A) Sum of abscissae of points of contact is 0 (B) Sum of abscissae of points of contact is 4
 (C) Sum of ordinates of points of contact is 0 (D) Sum of ordinates of points of contact is 4
TN0090
14. A tangent to the curve $x^2 + y^2 = 4$ at the point $(1, \sqrt{3})$ cut intercepts p and q with x and y -axis respectively, then-
 (A) $p = 4$ (B) $p = \frac{4}{\sqrt{3}}$ (C) $p - \sqrt{3}q = 0$ (D) $p^2 + q^2 = 32$
TN0091
15. If the curve $y^2 = 4ax$, $a > 0$ cuts the curve $xy = b$, $b > 0$ at right angle, then -
 (A) $a = \frac{1}{\sqrt{2}}, b = \sqrt{2}$ (B) $a = \frac{1}{2}, b = \sqrt{2}$ (C) $a = \frac{1}{4}, b = \frac{1}{2\sqrt{2}}$ (D) $a = \frac{1}{4}, b = \sqrt{2}$
TN0092

EXERCISE (O-3)

Linked Comprehension Type

Paragraph for Question 1 to 3

Let $P(h, k)$ be any point on curve $y = f(x)$. Let tangent drawn at point P meets x -axis at T & normal at point P meets x -axis at N (as shown in figure) and $m = \left. \frac{dy}{dx} \right|_{(h, k)}$ = slope of tangent.



- (i) Length of Tangent = $PT = |k| \sqrt{1 + \frac{1}{m^2}}$
 (ii) Length of Normal = $PN = |k| \sqrt{1 + m^2}$
 (iii) Length of subtangent = Projection of segment PT on x -axis = $TM = \left| \frac{k}{m} \right|$
 (iv) Length of subnormal = projection of line segment PN on x axis = $MN = |km|$

1. The product of length of tangent and length of normal for the curve $y = x^3 + 3x^2 + 4x - 1$ at point $x = 0$ is **TN0093**
- (A) $\frac{17}{4}$ (B) $\frac{\sqrt{15}}{4}$ (C) 17 (D) $\frac{4}{\sqrt{17}}$
2. Value of 'p' such that the length of the subtangent and subnormal is equal for the curve $y = e^{px} + px$ at the point (0, 1) **TN0094**
- (A) ± 1 (B) ± 2 (C) $\pm \frac{1}{2}$ (D) $\pm \frac{1}{4}$
3. The length of subnormal to $x = \sqrt{2} \cos t$, $y = -3 \sin t$ at $t = \frac{-\pi}{4}$. **TN0095**
- (A) $\frac{2}{9}$ (B) 1 (C) $\frac{7}{2}$ (D) $\frac{9}{2}$

Paragraph for Question 4 and 5

Water is flowing out at the rate of $6 \text{ m}^3/\text{min}$ from a reservoir shaped like a hemispherical bowl of radius $R = 13 \text{ m}$. The volume of water in the hemispherical bowl is given by $V = \frac{\pi}{3} \cdot y^2 (3R - y)$ when the water is y meter deep. Find

4. At what rate is the water level changing when the water is 8 m deep. **TN0044**
- (A) $\frac{-1}{24\pi} \text{ m/min.}$ (B) $\frac{-5}{288\pi} \text{ m/min.}$ (C) $\frac{-1}{12\pi} \text{ m/min.}$ (D) $\frac{-5}{144\pi} \text{ m/min.}$
5. At what rate is the radius of the water surface changing when the water is 8 m deep. **TN0044**
- (A) $\frac{-1}{24\pi} \text{ m/min.}$ (B) $\frac{-5}{288\pi} \text{ m/min.}$ (C) $\frac{-1}{12\pi} \text{ m/min.}$ (D) $\frac{-5}{144\pi} \text{ m/min.}$

Paragraph for Question 6 and 7

Consider a man 1.5 m tall walking on the pavement away from a lamp post 4.5 m high at the rate of 4 km/hr, then answer the following

6. How fast is the farther end of the shadow moving on the pavement (in km/hr) **TN0035**
7. How fast is his shadow lengthening (in km/hr) **TN0035**

Matrix Match Type

8. Consider the curve $y = e^{2x^2} + 2x$, then match List-I with List-II and select the correct answer using the code given below the list. TN0096

List-I

List-II

(P) Length of tangent at $x = 0$ is

(1) $\sqrt{5}$

(Q) Length of normal at $x = 0$ is

(2) $\frac{1}{2}$

(R) Length of subtangent at $x = 0$ is

(3) $\frac{\sqrt{5}}{2}$

(S) Length of subnormal at $x = 0$ is

(4) 2

(A) $P \rightarrow 1; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 4$

(B) $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 2$

(C) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 4$

(D) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 2$

9. Column - I

Column - II

(A) If curves $y^2 = 4ax$ and $y = e^{\frac{-x}{2a}}$ are orthogonal then 'a' can take value

(P) 3

(B) If θ is angle between the curves $y = [|\sin x| + |\cos x|]$, ($[\cdot]$ denote GIF) and $x^2 + y^2 = 5$ then $\operatorname{cosec}^2 \theta$ is

(Q) 1

(C) If curves $y^2 = 4a(x + a)$ and $y^2 = 4b(x + b)$ intersects

(R) $5/4$

each other orthogonally then $\left| \frac{a}{b} \right|$ can be equal to _____

(D) If $y = x^2 + 3x + c$ and $x = y^2 + 3y + c$ touches each other at (h, k) then $|h + k + c|$ can be equal to.....

(S) 2

TN0097

10. Column-I

Column-II

(A) The slope of the curve $2y^2 = ax^2 + b$ at $(1, -1)$ is -1 , then

(p) $a - b = 2$

(B) If (a, b) be the point on the curve $9y^2 = x^3$ where normal to the curve makes equal intercepts with the axes, then

(q) $a - b = 7/2$

(C) If the tangent at a point $(1, 2)$ on the curve $y = ax^2 + bx + \frac{7}{2}$ be parallel to the normal at $(-2, 2)$ on the curve $y = x^2 + 6x + 10$, then

(r) $a - b = 4/3$

(D) If the tangent to the curve $xy + ax + by = 0$ at $(1, 1)$ is inclined at an angle $\tan^{-1} 2$ with x-axis, then

(s) $a - b = 3$

TN0098

EXERCISE (O-4)**Numerical Grid Type**

1. Water is being poured on to a cylindrical vessel at the rate of $1 \text{ m}^3/\text{min}$ and the vessel has a circular base of radius 3 m. If the rate at which the level of water is rising in the vessel is $\frac{1}{k\pi} \text{ m/min}$, then the value of 'k' is
TN0034
2. An inverted cone has a depth of 10 cm & a base of radius 5 cm. Water is poured into it at the rate of $1.5 \text{ cm}^3/\text{min}$. If the rate at which level of water in the cone is rising, when the depth of water is 4 cm is $\frac{3}{k\pi} \text{ cm/min}$, then the value of 'k' is
TN0037
3. Water is dripping out from a conical funnel of semi vertical angle $\pi/4$, at the uniform rate of $2 \text{ cm}^3/\text{sec}$ through a tiny hole at the vertex at the bottom. When the slant height of the water is 4 cm, then the rate of decrease of the slant height of the water is given by $\frac{\sqrt{2}}{k\pi} \text{ cm/sec}$, then the value of 'k' is
TN0038
4. A water tank has the shape of a right circular cone with its vertex down. Its altitude is 10 cm and the radius of the base is 15 cm. Water leaks out of the bottom at a constant rate of 1 cu. cm/sec . Water is poured into the tank at a constant rate of $C \text{ cu. cm/sec}$. Compute C so that the water level will be rising at the rate of 4 cm/sec at the instant when the water is 2 cm deep. If C is of the form $p + q\pi$, then the value of $p + q$ is
TN0041
5. Sand is pouring from a pipe at the rate of 12 cc/sec . The falling sand forms a cone on the ground in such a way that the height of the cone is always $1/6$ th of the radius of the base. If the height of the sand cone increase at a rate $\frac{1}{p\pi}$ when the height is 4 cm, then the value of 'p' is
TN0042
6. A circular ink blot grows at the rate of 2 cm^2 per second. Then the rate at which the radius is increasing after $2\frac{6}{11}$ seconds is (Use $\pi = \frac{22}{7}$)
TN0043

7. The tangent to $y = ax^2 + bx + \frac{7}{2}$ at $(1, 2)$ is parallel to the normal at the point $(-2, 2)$ on the curve $y = x^2 + 6x + 10$. Then value of $a - 2b$ is

TN0049

8. A variable ΔABC in the xy plane has its orthocentre at vertex 'B', a fixed vertex 'A' at the origin and the third vertex 'C' restricted to lie on the parabola $y = 1 + \frac{7x^2}{36}$. The point B starts at the point $(0, 1)$ at time $t = 0$ and moves upward along the y axis at a constant velocity of 2 cm/sec, then how fast is the area of the triangle increasing when $t = \frac{7}{2}$ sec.

TN0059

9. There is a point (p, q) on the graph of $f(x) = x^2$ and a point (r, s) on the graph of $g(x) = -8/x$ where $p > 0$ and $r > 0$. If the line through (p, q) and (r, s) is also tangent to both the curves at these points respectively then the value of $(p + q)$ is

TN0063

10. The chord of the parabola $y = -a^2x^2 + 5ax - 4$ touches the curve $y = \frac{1}{1-x}$ at the point $x = 2$ and is bisected by that point, then the value of 'a' is

TN0065

EXERCISE (JM)

1. The normal to the curve, $x^2 + 2xy - 3y^2 = 0$, at $(1, 1)$: [JEE-MAIN 2015]

- (1) meets the curve again in the third quadrant
 (2) meets the curve again in the fourth quadrant
 (3) does not meet the curve again
 (4) meets the curve again in the second quadrant

TN0069

2. Consider $f(x) = \tan^{-1}\left(\sqrt{\frac{1+\sin x}{1-\sin x}}\right)$, $x \in \left(0, \frac{\pi}{2}\right)$. A normal to $y = f(x)$ at $x = \frac{\pi}{6}$ also passes through the point : [JEE-MAIN 2016]

- (1) $\left(\frac{\pi}{4}, 0\right)$ (2) $(0, 0)$ (3) $\left(0, \frac{2\pi}{3}\right)$ (4) $\left(\frac{\pi}{6}, 0\right)$

TN0070

3. The normal to the curve $y(x-2)(x-3) = x+6$ at the point where the curve intersects the y-axis passes through the point : [JEE-MAIN 2017]

- (1) $\left(\frac{1}{2}, \frac{1}{3}\right)$ (2) $\left(-\frac{1}{2}, -\frac{1}{2}\right)$ (3) $\left(\frac{1}{2}, \frac{1}{2}\right)$ (4) $\left(\frac{1}{2}, -\frac{1}{3}\right)$

TN0071

4. If the curves $y^2 = 6x$, $9x^2 + by^2 = 16$ intersect each other at right angles, then the value of b is :

- (1) $\frac{7}{2}$ (2) 4 (3) $\frac{9}{2}$ (4) 6 [JEE-MAIN 2018]

TN0072

5. The tangent to the curve, $y = xe^{x^2}$ passing through the point $(1, e)$ also passes through the point :

[JEE-MAIN 2019]

- (1) $\left(\frac{4}{3}, 2e\right)$ (2) $(2, 3e)$ (3) $\left(\frac{5}{3}, 2e\right)$ (4) $(3, 6e)$

TN0073

6. A 2 m ladder leans against a vertical wall. If the top of the ladder begins to slide down the wall at the rate 25 cm/sec., then the rate (in cm/sec.) at which the bottom of the ladder slides away from the wall on the horizontal ground when the top of the ladder is 1 m above the ground is :

[JEE-MAIN 2019]

- (1) $25\sqrt{3}$ (2) 25 (3) $\frac{25}{\sqrt{3}}$ (4) $\frac{25}{3}$

TN0099

7. The length of the perpendicular from the origin, on the normal to the curve, $x^2 + 2xy - 3y^2 = 0$ at the point $(2, 2)$ is [JEE-MAIN 2020]

- (1) $4\sqrt{2}$ (2) $2\sqrt{2}$ (3) 2 (4) $\sqrt{2}$

TN0074

8. Let the normal at a point P on the curve $y^2 - 3x^2 + y + 10 = 0$ intersect the y-axis at $\left(0, \frac{3}{2}\right)$. If m is the slope of the tangent at P to the curve, then |m| is equal to [JEE-MAIN 2020]

TN0075

9. Let P(h, k) be a point on the curve $y = x^2 + 7x + 2$, nearest to the line, $y = 3x - 3$. Then the equation of the normal to the curve at P is : [JEE-MAIN 2020]

(1) $x + 3y - 62 = 0$ (2) $x - 3y - 11 = 0$ (3) $x - 3y + 22 = 0$ (4) $x + 3y + 26 = 0$

TN0100

10. The equation of the normal to the curve $y = (1+x)^{2y} + \cos^2(\sin^{-1}x)$ at $x = 0$ is : [JEE-MAIN 2020]

(1) $y = 4x + 2$ (2) $x + 4y = 8$ (3) $y + 4x = 2$ (4) $2y + x = 4$

TN0101

11. Which of the following points lies on the tangent to the curve $x^4 e^y + 2\sqrt{y+1} = 3$ at the point (1, 0) ?

(1) (2, 2) (2) (-2, 6) (3) (-2, 4) (4) (2, 6)

[JEE-MAIN 2020]

TN0102

12. If the tangent to the curve, $y = f(x) = x \log_e x$, ($x > 0$) at a point (c, f(c)) is parallel to the line - segment joining the points (1, 0) and (e, e), then c is equal to : [JEE-MAIN 2021]

(1) $\frac{1}{e-1}$ (2) $e^{\left(\frac{1}{1-e}\right)}$ (3) $e^{\left(\frac{1}{e-1}\right)}$ (4) $\frac{e-1}{e}$

TN0103

13. An angle of intersection of the curves, $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and $x^2 + y^2 = ab$, $a > b$, is : [JEE-MAIN 2021]

(1) $\tan^{-1}\left(\frac{a+b}{\sqrt{ab}}\right)$ (2) $\tan^{-1}\left(\frac{a-b}{2\sqrt{ab}}\right)$ (3) $\tan^{-1}\left(\frac{a-b}{\sqrt{ab}}\right)$ (4) $\tan^{-1}(2\sqrt{ab})$

TN0104

14. If the tangent to the curve $y = x^3$ at the point P(t, t³) meets the curve again at Q, then the ordinate of the point which divides PQ internally in the ratio 1 : 2 is : [JEE-MAIN 2022]

(1) $-2t^3$ (2) 0 (3) $-t^3$ (4) $2t^3$

TN0105

15. The surface area of a balloon of spherical shape being inflated, increases at a constant rate. If initially, the radius of balloon is 3 units and after 5 seconds, it becomes 7 units, then its radius after 9 seconds is : [JEE-MAIN 2022]

(A) 9 (B) 10 (C) 11 (D) 12

TN0106

16. The tangent at the point (x_1, y_1) on the curve $y = x^3 + 3x^2 + 5$ passes through the origin, then (x_1, y_1) does NOT lie on the curve : [JEE-MAIN 2022]

(A) $x^2 + \frac{y^2}{81} = 2$ (B) $\frac{y^2}{9} - x^2 = 8$ (C) $y = 4x^2 + 5$ (D) $\frac{x}{3} - y^2 = 2$

TN0107

17. Water is being filled at the rate of $1 \text{ cm}^3 / \text{sec}$ in a right circular conical vessel (vertex downwards) of height 35 cm and diameter 14 cm. When the height of the water level is 10 cm, the rate (in cm^2 / sec) at which the wet conical surface area of the vessel increases is [JEE-MAIN 2022]

(A) 5 (B) $\frac{\sqrt{21}}{5}$ (C) $\frac{\sqrt{26}}{5}$ (D) $\frac{\sqrt{26}}{10}$

TN0108

18. If the angle made by the tangent at the point (x_0, y_0) on the curve $x = 12(t + \sin t \cos t)$, $y = 12(1 + \sin t)^2$, $0 < t < \frac{\pi}{2}$, with the positive x-axis is $\frac{\pi}{3}$, then y_0 is equal to [JEE-MAIN 2022]

(A) $6(3 + 2\sqrt{2})$ (B) $3(7 + 4\sqrt{3})$ (C) 27 (D) 48

TN0109

19. A water tank has the shape of a right circular cone with axis vertical and vertex downwards. Its semi-vertical angle is $\tan^{-1} \frac{3}{4}$. Water is poured in it at a constant rate of 6 cubic meter per hour. The rate (in square meter per hour), at which the wet curved surface area of the tank is increasing, when the depth of water in the tank is 4 meters, is _____. [JEE-MAIN 2022]

TN0110

EXERCISE (JA)

1. Find the equation of the straight line which is tangent at one point and normal at another point of the curve, $x = 3t^2$, $y = 2t^3$. **[REE 2000 (Mains) 5 out of 100]**

TN0076

2. If the normal to the curve, $y = f(x)$ at the point $(3, 4)$ makes an angle $\frac{3\pi}{4}$ with the positive x-axis. Then $f'(3) =$

- (A) -1 (B) $-\frac{3}{4}$ (C) $\frac{4}{3}$ (D) 1

[JEE 2000 (Scr.) 1 out of 35]

TN0077

3. The point(s) on the curve $y^3 + 3x^2 = 12y$ where the tangent is vertical, is(are)

- (A) $\left(\pm \frac{4}{\sqrt{3}}, -2\right)$ (B) $\left(\pm \sqrt{\frac{11}{3}}, 1\right)$ (C) $(0, 0)$ (D) $\left(\pm \frac{4}{\sqrt{3}}, 2\right)$

[JEE 2002 (Scr.), 3]

TN0078

4. Tangent to the curve $y = x^2 + 6$ at a point P (1, 7) touches the circle $x^2 + y^2 + 16x + 12y + c = 0$ at a point Q. Then the coordinates of Q are

- (A) $(-6, -11)$ (B) $(-9, -13)$ (C) $(-10, -15)$ (D) $(-6, -7)$

[JEE 2005 (Scr.), 3]

TN0079

5. The tangent to the curve $y = e^x$ drawn at the point (c, e^c) intersects the line joining the points $(c - 1, e^{c-1})$ and $(c + 1, e^{c+1})$

- (A) on the left of $x = c$ (B) on the right of $x = c$
(C) at no point (D) at all points

[JEE 2007, 3]

TN0080

ANSWER KEY

Do yourself - 1

- | | | | |
|---|-----------------------------------|---------------------------------------|------------------------------------|
| 1. 12π cm | 2. $80 \pi \text{ cm}^2/\text{s}$ | 3. (i) $10\pi \text{ cm}^2/\text{cm}$ | (ii) $5\pi \text{ cm}^2/\text{cm}$ |
| 4. $\frac{4}{\pi} \text{ cm}^2/\text{sec.}$ | 5. $7 \text{ cm}^2/\text{sec.}$ | 6. $\frac{1}{48\pi} \text{ cm/sec.}$ | 7. C |
| 8. A | 9. C | 10. B | |

Do yourself - 2

- | | | | |
|--------------|---------------------|---------------------------|------|
| 1. 0.208 | 2. 7.035 | 3. 2.00187 | 4. A |
| 5. (a) 6.05, | (b) $\frac{80}{27}$ | 6. $9.72\pi \text{ cm}^3$ | |

Do yourself - 3

- | | | | |
|--|--------------------|-------|-------|
| 1. $\frac{2}{\sqrt{5}}$ units | 2. $x - y + 5 = 0$ | 3. B | |
| 4. Tangent : $\left(y - \frac{a}{5}\right) = \frac{13}{16}\left(x - \frac{2a}{5}\right)$; Normal : $\left(y - \frac{a}{5}\right) = -\frac{16}{13}\left(x - \frac{2a}{5}\right)$ | | | |
| 5. D | 6. C | 7. C | 8. A |
| 9. C | 10. C | 11. C | 12. D |

Do yourself - 4

- | | | |
|--|--|------|
| 1. $\left \frac{\ell \ln a - \ell \ln b}{1 + \ell \ln a \ell \ln b} \right $ | 2. $\tan^{-1}\left(\frac{4\sqrt{2}}{7}\right)$ | 3. C |
| 4. But at (0,0) clearly x-axis & y-axis are their respective tangents hence they are orthogonal at (0,0) but not at (1,1). Hence these curves are not said to be orthogonal. | | |
| 5. B | | |

Do yourself - 5

- | | | | |
|------|------|------|------|
| 2. C | 3. C | 4. C | 5. A |
| 6. D | | | |

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	D	A	B	A	C	C	C	B	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	D	C	D	B	D	C	A	D	D
Que.	21	22	23	24	25					
Ans.	C	D	D	B	D					

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,D	A,B	B,C	B,C,D	B,D	C,D	A,B,C	B,C	A,B,C,D	A,C
Que.	11	12	13	14	15					
Ans.	A,B,C,D	A,B,C	B,D	A,C	B,C					

EXERCISE # O-3

Que.	1	2	3	4	5	6	7	8		
Ans.	A	C	D	A	B	6.00	2.00	C		
Q.9	A	B	C	D	Q.10	A	B	C	D	
	P,Q,R,S	R	P,Q,R,S	Q		p	r	q	s	

EXERCISE # O-4

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	9	8	4	37	48	0.25	6.00	9.42 or 9.43	2.00	1.00

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	3	3	3	1	3	2	4.00	4	2
Que.	11	12	13	14	15	16	17	18	19	
Ans.	2	3	3	1	A	D	C	C	5	

EXERCISE # JEE-ADVANCED

1. $\sqrt{2}x + y - 2\sqrt{2} = 0$ or $\sqrt{2}x - y - 2\sqrt{2} = 0$

2. D

3. D

4. D

5. A

MONOTONICITY

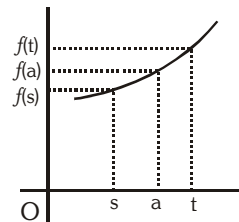
1. INCREASING / DECREASING / STRICTLY INCREASING / STRICTLY DECREASING NATURE OF A FUNCTION AT A POINT :

I. Increasing at $x = a$:

If $f(s) \leq f(a) \leq f(t)$ when ever $s < a < t$, where $s, t \in (a - h, a + h) \cap D_f$ for some $h > 0$, then f is said to be increasing at $x = a$.

- (1) When 'a' be left end of the interval
 $f(a) \leq f(x) \forall x \in (a, a + h) \cap D_f$ for some $h > 0$
 $\Rightarrow f$ is increasing at $x = a$.

- (2) When 'a' be right end of the interval
 $f(x) \leq f(a) \forall x \in (a - h, a) \cap D_f$ for some $h > 0$
 $\Rightarrow f$ is increasing at $x = a$.

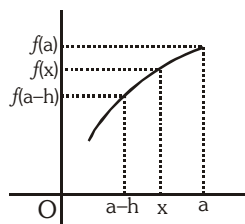
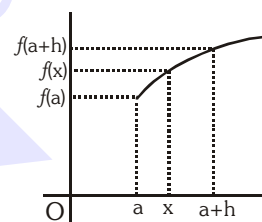


II. Strictly increasing at $x = a$:

If $f(s) < f(a) < f(t)$ when ever $s < a < t$, where $s, t \in (a - h, a + h) \cap D_f$ for some $h > 0$, then f is said to be strictly increasing at $x = a$.

- (1) When 'a' be left end of the interval
 $f(a) < f(x) \forall x \in (a, a + h) \cap D_f$ for some $h > 0$
 $\Rightarrow f$ is strictly increasing at $x = a$.

- (2) When 'a' be right end of the interval
 $f(x) < f(a) \forall x \in (a - h, a) \cap D_f$ for some $h > 0$
 $\Rightarrow f$ is strictly increasing at $x = a$.



III. Decreasing at $x = a$:

If $f(s) \geq f(a) \geq f(t)$ when ever $s < a < t$, where $s, t \in (a - h, a + h) \cap D_f$ for some $h > 0$, then f is said to be decreasing at $x = a$.

- (1) When 'a' be left end of the interval
 $f(a) \geq f(x) \forall x \in (a, a + h) \cap D_f$ for some $h > 0$
 $\Rightarrow f$ is decreasing at $x = a$.

- (2) When 'a' be right end of the interval
 $f(x) \geq f(a) \forall x \in (a - h, a) \cap D_f$ for some $h > 0$
 $\Rightarrow f$ is decreasing at $x = a$.

IV. Strictly decreasing at $x = a$:

If $f(s) > f(a) > f(t)$ when ever $s < a < t$, where $s, t \in (a - h, a + h) \cap D_f$ for some $h > 0$, then f is said to be strictly decreasing at $x = a$.

- (1) When 'a' be left end of the interval
 $f(a) > f(x) \forall x \in (a, a + h) \cap D_f$ for some $h > 0$
 $\Rightarrow f$ is strictly decreasing at $x = a$.

- (2) When 'a' be right end of the interval
 $f(x) > f(a) \forall x \in (a - h, a) \cap D_f$ for some $h > 0$
 $\Rightarrow f$ is strictly decreasing at $x = a$.

EXERCISE (O-1)

Straight Objective Type

- Let $f(x)$ and $g(x)$ be two continuous functions defined from $\mathbb{R} \rightarrow \mathbb{R}$, such that $f(x_1) > f(x_2)$ and $g(x_1) < g(x_2)$, $\forall x_1 > x_2$, then solution set of $f(g(\alpha^2 - 2\alpha)) > f(g(3\alpha - 4))$ is
(A) $\mathbb{R}$ (B) $\emptyset$ (C) $(1, 4)$ (D) $\mathbb{R} - [1, 4]$
MN0001
- When $0 \leq x \leq 1$, $f(x) = |x| + |x - 1|$ is -
(A) strictly increasing (B) strictly decreasing (C) constant (D) None of these
MN0002
- Complete set of values of K in order that $f(x) = \sin x - \cos x - Kx + b$ decreases for all real values is given by -
(A) $K < 1$ (B) $K \geq 1$ (C) $K \geq \sqrt{2}$ (D) $K < \sqrt{2}$
MN0003
- Function $f(x) = x^2(x - 2)^2$ is -
(A) increasing in $[0, 1]; [2, \infty)$ (B) decreasing in $[0, 1]; [2, \infty]$
(C) decreasing function (D) increasing function
MN0004
- If f and g are two strictly decreasing function such that $f \circ g$ is defined, then $f \circ g$ will be -
(A) strictly increasing function (B) strictly decreasing function
(C) neither increasing nor decreasing (D) None of these
MN0005
- If the function $f(x) = 2x^2 - kx + 5$ is strictly increasing in $[1, 2]$, then complete set of values of k is
(A) $(-\infty, 4]$ (B) $(4, \infty)$ (C) $(-\infty, 8]$ (D) $(8, \infty)$
MN0006
- The function x^x strictly decreases on the interval -
(A) $(0, e)$ (B) $(0, 1)$ (C) $\left(0, \frac{1}{e}\right)$ (D) None of these
MN0007
- If function $f(x) = 2x^2 + 3x - m \log x$ is monotonic decreasing in the interval $(0, 1)$, then the least value of the parameter m is -
(A) 7 (B) $\frac{15}{2}$ (C) $\frac{31}{4}$ (D) 8
MN0008

9. If $f(x) = x^3 - 10x^2 + 200x - 10$, then $f(x)$ is-

- (A) strictly decreasing in $(-\infty, 10]$ and strictly increasing in $[10, \infty)$
 (B) strictly increasing in $(-\infty, 10]$ and strictly decreasing in $[10, \infty)$
 (C) strictly increasing for every value of x
 (D) strictly decreasing for every value of x

MN0009

10. Which one of the following statements does not hold good for the function $f(x) = \cos^{-1}(2x^2 - 1)$?

- (A) f is not differentiable at $x = 0$ (B) f is monotonic
 (C) f is even (D) f has an extremum

MN0010

11. The function $f(x) = \tan^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ is -

- (A) strictly increasing in its domain
 (B) strictly decreasing in its domain
 (C) strictly decreasing in $(-\infty, 0)$ and strictly increasing in $(0, \infty)$
 (D) strictly increasing in $(-\infty, 0)$ and strictly decreasing in $(0, \infty)$

MN0011

12. The function $f : [a, \infty) \rightarrow \mathbb{R}$ where $\mathbb{R}$ denotes the range corresponding to the given domain, with rule $f(x) = 2x^3 - 3x^2 + 6$, will have an inverse provided

- (A) $a \geq 1$ (B) $a \geq 0$ (C) $a \leq 0$ (D) $a \leq 1$

MN0012

13. If $2a + 3b + 6c = 0$, then at least one root of the equation $ax^2 + bx + c = 0$ lies in the interval-

- (A) $(0, 1)$ (B) $(1, 2)$ (C) $(2, 3)$ (D) none

MN0013

14. If the equation $a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x = 0$ has a positive root $x = \alpha$, then the equation $na_n x^{n-1} + (n-1)a_{n-1} x^{n-2} + \dots + a_1 = 0$ has a positive root, which is-

- (A) Smaller than α (B) Greater than α
 (C) Equal to α (D) Greater than or equal to α

MN0014

15. A value of c for which the conclusion of Mean values theorem holds for the function $f(x) = \log_e x$ on the interval $[1, 3]$ is-

- (A) $2\log_3 e$ (B) $\frac{1}{2} \log_e 3$ (C) $\log_3 e$ (D) $\log_e 3$

MN0015

16. The value of c in Lagrange's theorem for the function $f(x) = \begin{cases} x \cos\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$ in the interval

$[-1, 1]$ is-

- (A) 0 (B) $\frac{1}{2}$
(C) $-\frac{1}{2}$ (D) Non existent in the interval

MN0016

17. If the function $f(x) = x^3 - 6x^2 + ax + b$ defined on $[1, 3]$, satisfies the Rolle's theorem for $c = \frac{2\sqrt{3}+1}{\sqrt{3}}$, then-
- (A) $a = 11, b = 6$ (B) $a = -11, b = 6$ (C) $a = 11, b \in \mathbb{R}$ (D) None of these

MN0017

18. Given: $f(x) = 4 - \left(\frac{1}{2} - x\right)^{2/3}$ $g(x) = \begin{cases} \frac{\tan[x]}{x}, & x \neq 0 \\ 1, & x = 0 \end{cases}$

$$h(x) = \{x\} \quad k(x) = 5^{\log_2(x+3)}$$

then in $[0, 1]$, Lagrange's Mean Value Theorem is NOT applicable to

- (A) f, g, h (B) h, k (C) f, g (D) g, h, k

where $[x]$ and $\{x\}$ denotes the greatest integer and fractional part function.

MN0018

19. Given $f'(1) = 1$ and $\frac{d}{dx}(f(2x)) = f'(x) \forall x > 0$. If $f'(x)$ is differentiable then there exists a number $c \in (2, 4)$ such that $f''(c)$ equals

- (A) $-1/4$ (B) $-1/8$ (C) $1/4$ (D) $1/8$

MN0019

20. If the function $f(x) = 2x^2 + 3x + 5$ satisfies LMVT at $x = 2$ on the closed interval $[1, a]$, then the value of 'a' is equal to

- (A) 3 (B) 4 (C) 6 (D) 1

MN0020

21. If $f : \mathbb{R} \rightarrow \mathbb{R}^+$ be a differentiable function and $g(x) = e^{2x}(2f(x) - 3(f(x))^2 + 2(f(x))^3) \forall x \in \mathbb{R}$, then which of the following is/are always correct -

- (A) $g(x)$ is strictly increasing wherever $f(x)$ is strictly increasing
 (B) $g(x)$ is strictly increasing wherever $f(x)$ is strictly decreasing
 (C) $g(x)$ is strictly decreasing wherever $f(x)$ is strictly decreasing
 (D) $g(x)$ is strictly decreasing wherever $f(x)$ is strictly increasing

MN0021

22. The equation $\sin x + x \cos x = 0$ has at least one root in

- (A) $\left(-\frac{\pi}{2}, 0\right)$ (B) $(0, \pi)$ (C) $\left(\pi, \frac{3\pi}{2}\right)$ (D) $\left(0, \frac{\pi}{2}\right)$

MN0022

23. Let a, b, c, d are non-zero real numbers such that $6a + 4b + 3c + 3d = 0$, then the equation $ax^3 + bx^2 + cx + d = 0$ has

- (A) atleast one root in $[-2, 0]$ (B) atleast one root in $[0, 2]$
 (C) atleast two roots in $[-2, 2]$ (D) no root in $[-2, 2]$

MN0023

24. The function $f(x) = \tan^{-1}(\sin x + \cos x)$ is an increasing function in -

- (A) $(\pi/4, \pi/2)$ (B) $(-\pi/2, \pi/4)$ (C) $(0, \pi/2)$ (D) $(-\pi/2, \pi/2)$

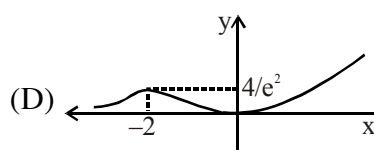
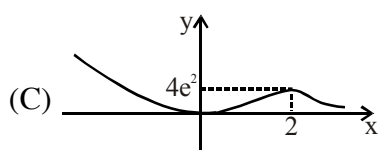
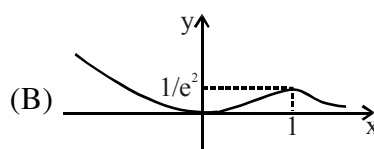
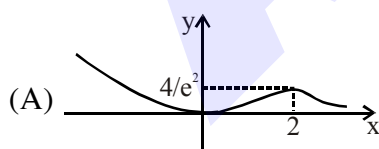
MN0062

25. The function $f(x) = (x - 3)^2$ satisfies all the conditions of Lagrange's mean value theorem in $[3, 4]$. A point on $y = (x - 3)^2$, where the tangent is parallel to the chord joining $(3, 0)$ and $(4, 1)$ is

- (A) $\left(\frac{7}{2}, \frac{1}{2}\right)$ (B) $\left(\frac{7}{2}, \frac{1}{4}\right)$ (C) $(1, 4)$ (D) $(4, 1)$

MN0094

26. Draw graph of $y = x^2 e^{-x}$.



MN0095

27. Let $f(x) = \begin{cases} \sin \frac{\pi x}{2}, & 0 \leq x < 1 \\ 3 - 2x, & x \geq 1 \end{cases}$ then :

- (A) $f(x)$ has local maxima at $x = 1$
- (B) $f(x)$ has local minima at $x = 1$
- (C) $f(x)$ does not have any local extrema at $x = 1$
- (D) $f(x)$ has a global minimum at $x = 1$

MN0096

28. Let $f(x) = \frac{x}{\sqrt{a^2 + x^2}} - \frac{d - x}{\sqrt{b^2 + (d - x)^2}}$, $x \in \mathbb{R}$, where a , b and d are non-zero real constants.

Then :-

- (A) f is a decreasing function of x
- (B) f is neither increasing nor decreasing function of x
- (C) f' is not a continuous function of x
- (D) f is an increasing function of x

MN0097

29. If $f : [1, 10] \Rightarrow [1, 10]$ is a non-decreasing function and $g : [1, 10] \Rightarrow [1, 10]$ is a non-increasing function. Let $h(x) = f(g(x))$ with $h(1) = 1$, then $h(2)$

- (A) lies in $(1, 2)$
- (B) is more than 2
- (C) is equal to 1
- (D) is not defined

MN0098

30. A function is matched below against an interval where it is supposed to be increasing. which of the following pairs is incorrectly matched ?

interval	function
(A) $(-\infty, \infty)$	$x^3 - 3x^2 + 3x + 3$
(B) $[2, \infty)$	$2x^3 - 3x^2 - 12x + 6$
(C) $\left(-\infty, \frac{1}{3}\right]$	$3x^2 - 2x + 1$
(D) $(-\infty, -4)$	$x^3 + 6x^2 + 6$

MN0061

EXERCISE (O-2)

Multiple Correct Answer Type

1. Which of the following is/are correct ?

- (A) Between any two roots of $e^x \cos x = 1$, there exists atleast one root of $\tan x = 1$.
(B) Between any two roots of $e^x \sin x = 1$, there exists atleast one root of $\tan x = -1$.
(C) Between any two roots of $e^x \cos x = 1$, there exists atleast one root of $e^x \sin x = 1$.
(D) Between any two roots of $e^x \sin x = 1$, there exists atleast one root of $e^x \cos x = -1$.

MN0024

2. Let $f(x) = \sin^2 x$ & $0 < A < B$, where $A, B \in \left(0, \frac{\pi}{4}\right)$, then

- (A) $(B - A) \sin 2A < \sin^2 B - \sin^2 A < (B - A) \sin 2B$
(B) $(B - A) \sin 2A > \sin^2 B - \sin^2 A > (B - A) \sin 2B$
(C) $\cos 2B < \cos 2A$
(D) $B \cos 2A > A \cos 2B$

MN0025

3. For the equation $\frac{e^{-x}}{x+1} = a$; which of the following statement(s) is/are correct ?

- (A) If $a \in (0, \infty)$, then equation has 2 real and distinct roots
(B) If $a \in (-\infty, -e^2)$, then equation has 2 real & distinct roots.
(C) If $a \in (0, \infty)$, then equation has 1 real root
(D) If $a \in (-e, 0)$, then equation has no real root.

MN0026

4. The number of roots of the equation $x^2 - 2x - \log_2 |1 - x| = 3$ is

- (A) 4 (B) 2 (C) 1 (D) 0

MN0027

5. Which of the following statements is/are correct ?

- (A) $x + \sin x$ is increasing function
(B) $\sec x$ is neither increasing nor decreasing function
(C) $x + \sin x$ is decreasing function
(D) $\sec x$ is an increasing function

MN0099

6. If $f(x) = 2x + \cot^{-1} x + \ln(\sqrt{1+x^2} - x)$, then $f(x)$:

(A) increases in $[0, \infty)$

(B) decreases in $[0, \infty)$

(C) neither increases nor decreases in $[0, \infty)$

(D) increases in $(-\infty, \infty)$

MN0100

7. Let $g(x) = 2f(x/2) + f(1 - x)$ and $f''(x) < 0$ in $0 \leq x \leq 1$ then $g(x)$ _____

(A) decreases in $\left[0, \frac{2}{3}\right]$

(B) decreases $\left[\frac{2}{3}, 1\right]$

(C) increases in $\left[0, \frac{2}{3}\right]$

(D) increases in $\left[\frac{2}{3}, 1\right]$

MN0101

8. Let $f(x) = x^{m/n}$ for $x \in \mathbb{R}$ where m and n are integers, m even and n odd and $0 < m < n$. Then

(A) $f(x)$ decreases on $(-\infty, 0]$

(B) $f(x)$ increases on $[0, \infty)$

(C) $f(x)$ increases on $(-\infty, 0]$

(D) $f(x)$ decreases on $[0, \infty)$

MN0102

9. Let $\phi(x) = (f(x))^3 - 3(f(x))^2 + 4f(x) + 5x + 3\sin x + 4\cos x \quad \forall x \in \mathbb{R}$, where $f(x)$ is a differentiable function $\forall x \in \mathbb{R}$, then

(A) ϕ is increasing whenever f is increasing

(B) ϕ is increasing whenever f is decreasing

(C) ϕ is decreasing whenever f is decreasing

(D) ϕ is decreasing if $f'(x) = -11$

MN0103

10. If p, q, r be real, then the intervals in which, $f(x) = \begin{vmatrix} x + p^2 & pq & pr \\ pq & x + q^2 & qr \\ pr & qr & x + r^2 \end{vmatrix}$,

(A) increase is $x < -\frac{2}{3}(p^2 + q^2 + r^2)$, $x > 0$

(B) decrease is $(-\frac{2}{3}(p^2 + q^2 + r^2), 0)$

(C) decrease is $x < -\frac{2}{3}(p^2 + q^2 + r^2)$, $x > 0$

(D) increase is $(-\frac{2}{3}(p^2 + q^2 + r^2), 0)$

MN0104

11. If $f(x) = \frac{x^2}{2-2\cos x}$; $g(x) = \frac{x^2}{6x-6\sin x}$ where $0 < x < 1$, then

(A) 'f' is increasing function

(B) 'g' is decreasing function

(C) $\frac{f(x)}{g(x)}$ is increasing function

(D) $g(f(x))$ is decreasing function

MN0105

12. For the function $f(x) = x \cot^{-1} x$, $x \geq 0$

- (A) there is atleast one $x \in (0, 1)$ for which $\cot^{-1} x = \frac{x}{1+x^2}$
- (B) for atleast one x in the interval $(0, \infty)$, $f\left(x + \frac{2}{\pi}\right) - f(x) < 1$
- (C) number of solution of the equation $f(x) = \sec x$ is 1
- (D) $f'(x)$ is strictly decreasing in the interval $(0, \infty)$

MN0106

13. Let $f(x)$ be a differentiable function and $f(\alpha) = f(\beta) = 0$ ($\alpha < \beta$), then in the interval (α, β)

- (A) $f(x) + f'(x) = 0$ has at least one root
- (B) $f(x) - f'(x) = 0$ has at least one real root
- (C) $f(x) \cdot f'(x) = 0$ has at least one real root
- (D) none of these

MN0107

14. Which of the following inequalities are valid -

- (A) $|\tan^{-1} x - \tan^{-1} y| \leq |x - y| \quad \forall x, y \in \mathbb{R}$
- (B) $|\tan^{-1} x - \tan^{-1} y| \geq |x - y|$
- (C) $|\sin x - \sin y| \leq |x - y|$
- (D) $|\sin x - \sin y| \geq |x - y|$

MN0108

15. For all x in $[1, 2]$

Let $f''(x)$ of a non-constant function $f(x)$ exist and satisfy $|f''(x)| \leq 2$. If $f(1) = f(2)$, then

- (A) There exist some $a \in (1, 2)$ such that $f'(a) = 0$
- (B) $f(x)$ is strictly increasing in $(1, 2)$
- (C) There exist atleast one $c \in (1, 2)$ such that $f'(c) > 0$
- (D) $|f'(x)| < 2 \quad \forall x \in [1, 2]$

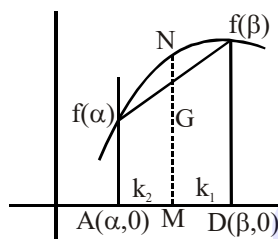
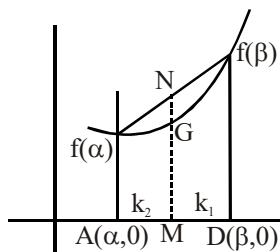
MN0109

EXERCISE (O-3)

Linked Comprehension Type

Paragraph for Question 1 to 3

For a double differentiable function $f(x)$ if $f''(x) \geq 0$ then $f(x)$ is concave upward and if $f''(x) \leq 0$ then $f(x)$ is concave downward



Here $M\left(\frac{k_1\alpha + k_2\beta}{k_1 + k_2}, 0\right)$

If $f(x)$ is a concave upward in $[a, b]$ and $\alpha, \beta \in [a, b]$ then $\frac{k_1 f(\alpha) + k_2 f(\beta)}{k_1 + k_2} \geq f\left(\frac{k_1\alpha + k_2\beta}{k_1 + k_2}\right)$, where $k_1, k_2 \in \mathbb{R}^+$

If $f(x)$ is a concave downward in $[a, b]$ and $\alpha, \beta \in [a, b]$ then $\frac{k_1 f(\alpha) + k_2 f(\beta)}{k_1 + k_2} \leq f\left(\frac{k_1\alpha + k_2\beta}{k_1 + k_2}\right)$, where $k_1, k_2 \in \mathbb{R}^+$

then answer the following :

1. Which of the following is true

- (A) $\frac{\sin \alpha + \sin \beta}{2} > \sin\left(\frac{\alpha + \beta}{2}\right)$; $\alpha, \beta \in (0, \pi)$
 (B) $\frac{\sin \alpha + \sin \beta}{2} < \sin\left(\frac{\alpha + \beta}{2}\right)$; $\alpha, \beta \in (\pi, 2\pi)$
 (C) $\frac{\sin \alpha + \sin \beta}{2} < \sin\left(\frac{\alpha + \beta}{2}\right)$; $\alpha, \beta \in (0, \pi)$
 (D) None of these

MN0110

2. Which of the following is true

- (A) $\frac{2^\alpha + 2^{\beta+1}}{3} \leq 2^{\frac{\alpha+2\beta}{3}}$ (B) $\frac{2\ln \alpha + \ln \beta}{3} \geq \ln\left(\frac{2\alpha + \beta}{3}\right)$
 (C) $\frac{\tan^{-1} \alpha + \tan^{-1} \beta}{2} \leq \tan^{-1}\left(\frac{\alpha + \beta}{2}\right)$; $a, b \in \mathbb{R}^-$ (D) $\frac{e^\alpha + 2e^\beta}{3} \geq e^{\frac{\alpha+2\beta}{3}}$

MN0111

3. Let α , β and γ are three distinct real numbers and $f''(x) < 0$. Also $f(x)$ is increasing function and

let $A = \frac{f^{-1}(\alpha) + f^{-1}(\beta) + f^{-1}(\gamma)}{3}$ and $B = f^{-1}\left(\frac{\alpha + \beta + \gamma}{3}\right)$, then order relation between A and B is ?

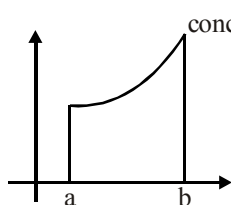
- (A) $A > B$ (B) $A < B$
(C) $A = B$ (D) none of these

MN0112

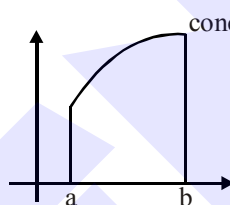
Paragraph for Question 4 and 5

If $f''(x) > 0 \forall x \in (a, b)$, then the curve $y = f(x)$ is concave up in (a, b) and

If $f''(x) < 0 \forall x \in (a, b)$ then the curve $y = f(x)$ is concave down in (a, b) .

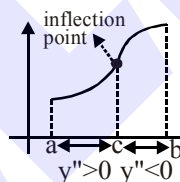


concave upward



concave downward

The point where concavity of the curve changes is known as point of inflection (at inflection point $f''(x)$ is equal to 0 or undefined).



4. Point(s) of inflection for $f(x) = (x - 1)^3 (x - 2)^2$, is/are

- (A) 1 (B) 2
(C) $\frac{16 + \sqrt{6}}{10}$ (D) $\frac{16 - \sqrt{6}}{10}$

MN0113

5. The value(s) of 'a' for which the function $f(x) = x^4 + ax^3 + \frac{3x^2}{2} + 1$ will be concave upward along the entire real line, is :

- (A) -3 (B) -2 (C) 1 (D) 2

MN0114

Paragraph for Question 6 to 8

If $f(x) = |x - 2| + |x - 4| + |x - 6|$.

6. If the set of values of x for which $f(x)$ increases is $[a + 2, \infty)$, then 'a' is - **MN0115**
7. If the set of values of x for which $f(x)$ decreases is $(-\infty, b - 1]$, then 'b' is - **MN0116**
8. If $f(x)$ is symmetrical about the line $x = K$, then K is equal to - **MN0117**

Matrix Match Type

9. Match List-I with List-II and select the correct answer using the code given below the list.

List - I

List - II

- | | | |
|-----------------------------------|---|---------------|
| (P) $f(x) = 2 \cdot e^{x^2 - 4x}$ | (1) Strictly increasing in $[2, \infty)$ &
Strictly decreasing in $(-\infty, 2]$ | MN0028 |
| (Q) $f(x) = e^x/x$ | (2) Strictly increasing in $[0, 2]$ &
Strictly decreasing in $(-\infty, 0] \cup [2, \infty)$ | MN0029 |
| (R) $f(x) = x^2 e^{-x}$ | (3) Strictly increasing in $[1, \infty)$ &
Strictly decreasing in $(-\infty, 0) \cup (0, 1]$ | MN0030 |
| (S) $f(x) = 2x^2 - \ln x $ | (4) Strictly increasing in $\left[-\frac{1}{2}, 0\right) \cup \left[\frac{1}{2}, \infty\right)$ &
Strictly decreasing in $\left(-\infty, -\frac{1}{2}\right] \cup \left(0, \frac{1}{2}\right]$ | MN0031 |

(A) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 1$

(B) $P \rightarrow 1; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 4$

(C) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 4; S \rightarrow 1$

(D) $P \rightarrow 4; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 2$

10. Column - I

Column - II

- | | |
|--|---|
| (A) $f(x) = \frac{\sin x}{e^x}, x \in [0, \pi]$ | (P) Conditions in Rolle's theorem are satisfied. |
| (B) $f(x) = \operatorname{sgn}((e^x - 1) \ln x), x \in \left[\frac{1}{2}, \frac{3}{2}\right]$ | (Q) Conditions in LMVT are satisfied. |
| (C) $f(x) = (x - 1)^{2/5}, x \in [0, 3]$ | (R) At least one condition in Rolle's theorem is not satisfied. |
| (D) $f(x) = \begin{cases} x \left(\frac{e^{\frac{1}{x}} - 1}{e^{\frac{1}{x}} + 1} \right) & ; x \in [-1, 1] - \{0\} \\ 0 & ; x = 0 \end{cases}$ | (S) At least one condition in LMVT is not satisfied. |

MN0118

EXERCISE (O-4)

Numerical Grid Type

1. If the sum of greatest & the least values of the function $f(x) = 12x^{4/3} - 6x^{1/3}$ in the interval $x \in [-1, 1]$, if they exist, is S then the value of $4S$ is MN0037
2. If $f(x) = 2e^x - ae^{-x} + (2a+1)x - 3$ increases for every $x \in \mathbb{R}$ then the range of values of 'a' is $[k, \infty)$ find k . MN0039
3. Find the sum of value of a , m and b if the function $f(x) = \begin{cases} 3 & x = 0 \\ -x^2 + 3x + a & 0 < x < 1 \\ mx + b & 1 \leq x \leq 2 \end{cases}$ satisfy the hypothesis of the mean value theorem for the interval $[0, 2]$. MN0049
4. Find the least integral value of 'a' for which the function $f(x) = \sin x - a \sin 2x - \frac{1}{3} \sin 3x + 2ax$ strictly increases throughout the number line. MN0051
5. Find the number of integral values of 'a' for which the function, $f(x) = \left(1 - \frac{\sqrt{21 - 4a - a^2}}{a + 1}\right)x^3 + 5x + \sqrt{7}$ is strictly increasing at every point of its domain. MN0052
6. Let $f(x)$ be a strictly increasing function defined on $(0, \infty)$. If $f(2a^2 + a + 1) > f(3a^2 - 4a + 1)$. The range of a is $\left(0, \frac{1}{p}\right) \cup (q, r)$. Find $(p + q + r)$. MN0053
7. Find the minimum value of the function $f(x) = x^{3/2} + x^{-3/2} - 4\left(x + \frac{1}{x}\right)$ for all permissible real x . MN0054
8. If f and g are differentiable functions in $[0, 1]$ satisfying $f(0) = 2 = g(1)$, $g(0) = 0$ and $f(1) = 6$, then $a.f'(c) = b.g'(c)$ for some $c \in]0, 1[$, then the value of $\frac{b}{a}$ is : MN0064
9. If $b > a$ and the minimum value of $|x - a|^3 + |x - b|^3$, $x \in \mathbb{R}$, is $\frac{(b-a)^p}{k}$, then the value of $(k - p)$ is MN0040
10. If the set of values of x for which the inequality $\ln(1+x) > x/(1+x)$ is valid is $(a, b) \cup (b, \infty)$, then the value of $(2a - b)$ is MN0041

EXERCISE (JM)

1. If the function f given by $f(x) = x^3 - 3(a-2)x^2 + 3ax + 7$, for some $a \in \mathbb{R}$ is increasing in $(0, 1]$ and decreasing in $[1, 5)$, then a root of the equation, $\frac{f(x)-14}{(x-1)^2} = 0 (x \neq 1)$ is : **[JEE-MAIN 2019]**

(1) 6 (2) 5 (3) 7 (4) -7

MN0065

2. Let $f(x) = e^x - x$ and $g(x) = x^2 - x$, $\forall x \in \mathbb{R}$. Then the set of all $x \in \mathbb{R}$, where the function $h(x) = (f \circ g)(x)$ is increasing, is : **[JEE-MAIN 2019]**

(1) $\left[-1, \frac{-1}{2}\right] \cup \left[\frac{1}{2}, \infty\right)$ (2) $\left[0, \frac{1}{2}\right] \cup [1, \infty)$ (3) $\left[\frac{-1}{2}, 0\right] \cup [1, \infty)$ (4) $[0, \infty)$

MN0066

3. If m is the minimum value of k for which the function $f(x) = x\sqrt{kx - x^2}$ is increasing in the interval $[0, 3]$ and M is the maximum value of f in $[0, 3]$ when $k = m$, then the ordered pair (m, M) is equal to : **[JEE-MAIN 2019]**

(1) $(4, 3\sqrt{2})$ (2) $(4, 3\sqrt{3})$ (3) $(3, 3\sqrt{3})$ (4) $(5, 3\sqrt{6})$

MN0067

4. The value of c in the Lagrange's mean value theorem for the function $f(x) = x^3 - 4x^2 + 8x + 11$, when $x \in [0, 1]$ is : **[JEE-MAIN 2020]**

(1) $\frac{2}{3}$ (2) $\frac{\sqrt{7}-2}{3}$ (3) $\frac{4-\sqrt{5}}{3}$ (4) $\frac{4-\sqrt{7}}{3}$

MN0068

5. Let the function, $f : [-7, 0] \rightarrow \mathbb{R}$ be continuous on $[-7, 0]$ and differentiable on $(-7, 0)$. If $f(-7) = -3$ and $f'(x) \leq 2$, for all $x \in (-7, 0)$, then for all such functions f , $f(-1) + f(0)$ lies in the interval : **[JEE-MAIN 2020]**

(1) $[-6, 20]$ (2) $(-\infty, 20]$ (3) $(-\infty, 11]$ (4) $[-3, 11]$

MN0069

6. Let $f(x) = x \cos^{-1}(-\sin|x|)$, $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, then which of the following is true ? **[JEE-MAIN 2020]**

(1) f' is decreasing in $\left(-\frac{\pi}{2}, 0\right)$ and increasing in $\left(0, \frac{\pi}{2}\right)$

(2) f is not differentiable at $x = 0$

(3) $f'(0) = -\frac{\pi}{2}$

(4) f' is increasing in $\left(-\frac{\pi}{2}, 0\right)$ and decreasing in $\left(0, \frac{\pi}{2}\right)$

MN0070

7. Let f be any function continuous on $[a, b]$ and twice differentiable on (a, b) . If for all $x \in (a, b)$, $f'(x) > 0$

and $f''(x) < 0$, then for any $c \in (a, b)$, $\frac{f(c) - f(a)}{f(b) - f(c)}$ is greater than: [JEE-MAIN 2020]

- (1) $\frac{b+a}{b-a}$ (2) $\frac{b-c}{c-a}$ (3) $\frac{c-a}{b-c}$ (4) 1

MN0071

8. Let $f : \mathbf{R} \rightarrow \mathbf{R}$ be defined as,

$$f(x) = \begin{cases} -55x, & \text{if } x < -5 \\ 2x^3 - 3x^2 - 120x, & \text{if } -5 \leq x \leq 4 \\ 2x^3 - 3x^2 - 36x - 336, & \text{if } x > 4, \end{cases}$$

Let $A = \{x \in \mathbf{R} : f \text{ is increasing}\}$. Then A is equal to :

- (1) $(-\infty, -5) \cup (4, \infty)$ (2) $(-5, \infty)$
(3) $(-\infty, -5) \cup (-4, \infty)$ (4) $(-5, -4) \cup (4, \infty)$ [JEE-MAIN 2021]

MN0119

9. If Rolle's theorem holds for the function $f(x) = x^3 - ax^2 + bx - 4$, $x \in [1, 2]$ with $f'\left(\frac{4}{3}\right) = 0$, then ordered

pair (a, b) is equal to :

- (1) $(5, 8)$ (2) $(-5, 8)$ (3) $(5, -8)$ (4) $(-5, -8)$ [JEE-MAIN 2021]

MN0120

10. Consider the function $f : \mathbf{R} \rightarrow \mathbf{R}$ defined by $f(x) = \begin{cases} \left(2 - \sin\left(\frac{1}{x}\right)\right)|x|, & x \neq 0 \\ 0, & x = 0 \end{cases}$. Then f is :

- (1) monotonic on $(-\infty, 0) \cup (0, \infty)$ (2) not monotonic on $(-\infty, 0)$ and $(0, \infty)$
(3) monotonic on $(0, \infty)$ only (4) monotonic on $(-\infty, 0)$ only

[JEE-MAIN 2021]

MN0121

11. Let $f : \mathbf{R} \rightarrow \mathbf{R}$ and $g : \mathbf{R} \rightarrow \mathbf{R}$ be two functions defined by $f(x) = \log_e(x^2 + 1) - e^{-x} + 1$ and $g(x) = \frac{1 - 2e^{2x}}{e^x}$.

Then, for which of the following range of α , the inequality $f\left(g\left(\frac{(\alpha-1)^2}{3}\right)\right) > f\left(g\left(\alpha - \frac{5}{3}\right)\right)$ holds ?

[JEE-MAIN 2022]

- (A) $(2, 3)$ (B) $(-2, -1)$ (C) $(1, 2)$ (D) $(-1, 1)$

MN0122

12. The number of real solutions of $x^7 + 5x^3 + 3x + 1 = 0$ is equal to _____. [JEE-MAIN 2022]

- (A) 0 (B) 1 (C) 3 (D) 5

MN0123

13. The function $f(x) = xe^{x(1-x)}$, $x \in \mathbb{R}$, is [JEE-MAIN 2022]

- (A) increasing in $\left(-\frac{1}{2}, 1\right)$ (B) decreasing in $\left(\frac{1}{2}, 2\right)$
 (C) increasing in $\left(-1, -\frac{1}{2}\right)$ (D) decreasing in $\left(-\frac{1}{2}, \frac{1}{2}\right)$

MN0124

EXERCISE (JA)

1. Let $f : (0,1) \rightarrow \mathbb{R}$ be defined by $f(x) = \frac{b-x}{1-bx}$, where b is a constant such that $0 < b < 1$. Then

[JEE 2011, 4M]

- (A) f is not invertible on $(0,1)$ (B) $f \neq f^{-1}$ on $(0,1)$ and $f'(b) = \frac{1}{f'(0)}$
 (C) $f = f^{-1}$ on $(0,1)$ and $f'(b) = \frac{1}{f'(0)}$ (D) f^{-1} is differentiable on $(0,1)$

MN0072

2. The number of distinct real roots of $x^4 - 4x^3 + 12x^2 + x - 1 = 0$ is [JEE 2011, 4M]

MN0073

3. The number of points in $(-\infty, \infty)$, for which $x^2 - x \sin x - \cos x = 0$, is [JEE 2013, 2M]

- (A) 6 (B) 4 (C) 2 (D) 0

MN0074

4. Let $f(x) = x \sin \pi x$, $x > 0$. Then for all natural numbers n , $f'(x)$ vanishes at - [JEE 2013, 4M, -1M]

- (A) a unique point in the interval $\left(n, n + \frac{1}{2}\right)$ (B) a unique point in the interval $\left(n + \frac{1}{2}, n + 1\right)$
 (C) a unique point in the interval $(n, n + 1)$ (D) two points in the interval $(n, n + 1)$

MN0075

5. Let $a \in \mathbb{R}$ and let $f : \mathbb{R} \rightarrow \mathbb{R}$ be given by $f(x) = x^5 - 5x + a$. Then [JEE(Advanced)-2014, 3]

- (A) $f(x)$ has three real roots if $a > 4$ (B) $f(x)$ has only one real roots if $a > 4$
 (C) $f(x)$ has three real roots if $a < -4$ (D) $f(x)$ has three real roots if $-4 < a < 4$

MN0076

6. Let $f : \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \rightarrow \mathbb{R}$ be given by $f(x) = (\log(\sec x + \tan x))^3$ Then :-

- (A) $f(x)$ is an odd function (B) $f(x)$ is a one-one function
 (C) $f(x)$ is an onto function (D) $f(x)$ is an even function

[JEE(Advanced)-2014, 3]

FN0195

7. Let $f, g : [-1, 2] \rightarrow \mathbb{R}$ be continuous function which are twice differentiable on the interval $(-1, 2)$. Let the values of f and g at the points $-1, 0$ and 2 be as given in the following table :

	$x = -1$	$x = 0$	$x = 2$
$f(x)$	3	6	0
$g(x)$	0	1	-1

[JEE 2015, 4M, -2M]

In each of the intervals $(-1, 0)$ and $(0, 2)$ the function $(f - 3g)''$ never vanishes. Then the correct statement(s) is(are)

- (A) $f'(x) - 3g'(x) = 0$ has exactly three solutions in $(-1, 0) \cup (0, 2)$
 (B) $f'(x) - 3g'(x) = 0$ has exactly one solution in $(-1, 0)$
 (C) $f'(x) - 3g'(x) = 0$ has exactly one solutions in $(0, 2)$
 (D) $f'(x) - 3g'(x) = 0$ has exactly two solutions in $(-1, 0)$ and exactly two solutions in $(0, 2)$

MN0077

Answer Q.8, Q.9 and Q.10 by appropriately matching the information given in the three columns of the following table.

[JEE(Advanced)-2017]

Let $f(x) = x + \log_e x - x \log_e x$, $x \in (0, \infty)$.

- * Column 1 contains information about zeros of $f(x)$, $f'(x)$ and $f''(x)$.
- * Column 2 contains information about the limiting behavior of $f(x)$, $f'(x)$ and $f''(x)$ at infinity.
- * Column 3 contains information about increasing/decreasing nature of $f(x)$ and $f'(x)$.

Column 1

Column 2

Column 3

- | | | |
|---|---|--------------------------------------|
| (I) $f(x) = 0$ for some $x \in (1, e^2)$ | (i) $\lim_{x \rightarrow \infty} f(x) = 0$ | (P) f is increasing in $(0, 1)$ |
| (II) $f'(x) = 0$ for some $x \in (1, e)$ | (ii) $\lim_{x \rightarrow \infty} f(x) = -\infty$ | (Q) f is decreasing in (e, e^2) |
| (III) $f'(x) = 0$ for some $x \in (0, 1)$ | (iii) $\lim_{x \rightarrow \infty} f'(x) = -\infty$ | (R) f' is increasing in $(0, 1)$ |
| (IV) $f''(x) = 0$ for some $x \in (1, e)$ | (iv) $\lim_{x \rightarrow \infty} f''(x) = 0$ | (S) f' is decreasing in (e, e^2) |

8. Which of the following options is the only **CORRECT** combination ?

- (A) (IV) (i) (S) (B) (I) (ii) (R) (C) (III) (iv) (P) (D) (II) (iii) (S)

MN0078

9. Which of the following options is the only **CORRECT** combination ?

- (A) (III) (iii) (R) (B) (I) (i) (P) (C) (IV) (iv) (S) (D) (II) (ii) (Q)

MN0078

10. Which of the following options is the only **INCORRECT** combination ?

- (A) (II) (iii) (P) (B) (II) (iv) (Q) (C) (I) (iii) (P) (D) (III) (i) (R)

MN0078

11. If $f : \mathbb{R} \rightarrow \mathbb{R}$ is a twice differentiable function such that $f''(x) > 0$ for all $x \in \mathbb{R}$, and

$$f\left(\frac{1}{2}\right) = \frac{1}{2}, f(1) = 1, \text{ then}$$

[JEE(Advanced)-2017]

- (A) $0 < f'(1) \leq \frac{1}{2}$ (B) $f'(1) \leq 0$ (C) $f'(1) > 1$ (D) $\frac{1}{2} < f'(1) \leq 1$

MN0079

12. For a polynomial $g(x)$ with real coefficient, let m_g denote the number of distinct real roots of $g(x)$. Suppose S is the set of polynomials with real coefficients defined by

$$S = \{(x^2 - 1)^2 (a_0 + a_1x + a_2x^2 + a_3x^3) : a_0, a_1, a_2, a_3 \in \mathbb{R}\}.$$

For a polynomial f , let f' and f'' denote its first and second order derivatives, respectively. Then the minimum possible value of $(m f' + m f'')$, where $f \in S$, is _____

[JEE(Advanced)-2020]

MN0092

13. $f : \mathbb{R} \rightarrow \mathbb{R}$ is defined by $f(x) = |x|(x - \sin x)$, then which of the following statements is **TRUE** ?

[JEE(Advanced)-2020]

- (A) f is one-one, but **NOT** onto (B) f is onto, but **NOT** one-one
(C) f is **BOTH** one-one and onto (D) f is **NEITHER** one-one **NOR** onto

FN0197

14. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by

$$f(x) = \frac{x^2 - 3x - 6}{x^2 + 2x + 4}.$$

Then which of the following statements is (are) TRUE ?

[JEE(Advanced)-2021]

- (A) f is decreasing in the interval $(-2, -1)$ (B) f is increasing in the interval $(1, 2)$
(C) f is onto (D) Range of f is $\left[-\frac{3}{2}, 2\right]$

MN0125

ANSWER KEY

Do yourself - 1

1. -4
2. Strictly increasing
4. Monotonically decreasing at $x = 0$ and Monotonically increasing at $x = 1$ & 2
6. D
7. A
8. D
9. D
10. A,B
11. C
12. D
13. A,B
14. A,D
15. A,D
16. (a) Strictly increasing in $[0, 3\pi/4]$; $[7\pi/4, 2\pi]$ & Strictly decreasing in $[3\pi/4, 7\pi/4]$
 (b) Strictly increasing in $[0, \pi/6]$; $[\pi/2, 5\pi/6]$; $[3\pi/2, 2\pi]$ & Strictly decreasing in $[\pi/6, \pi/2]$; $[5\pi/6, 3\pi/2]$
 (c) Strictly increasing in $[0, \pi/2] \cup [3\pi/2, 2\pi]$ and Strictly decreasing in $[\pi/2, 3\pi/2]$
17. $(-2, 0) \cup (2, \infty)$

Do yourself - 2

1. Greatest is $\frac{8}{3}$ and least value is $-\frac{8}{3}$
3. $\left[\frac{\pi}{3} - \frac{1}{2} \ln 3, \frac{\pi}{6} + \frac{1}{2} \ln 3 \right]$
5. C
6. D
7. A
8. A,B,C

Do yourself - 3

1. Rolles theorem is valid
2. (a) $f(x)$ is non-differentiable at $x = 0$ in $(-1, 1)$ (b) at $x = 1$, f is not defined
3. $a > \frac{1}{2}$
5. D
6. B
7. D
8. B
9. C

Do yourself - 4

2. $C = \sqrt{\frac{117}{9}}$
3. 2
4. 21
5. A
6. B
7. C
8. A
9. D

EXERCISE # 0-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	C	C	A	A	A	C	A	C	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	A	A	A	A	D	C	A	B	A
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	A	B	B	B	B	A	A	D	C	C

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B,C,D	A,C,D	B,C,D	A	A,B	A,D	B,C	A,B	A,D	A,B
Que.	11	12	13	14	15					
Ans.	A,B,C	B,D	A,B,C	A,C	A,C,D					

EXERCISE # O-3

Que.	1	2	3	4	5	6	7	8	9	
Ans.	C	D	A	A,C,D	B,C,D	2.00	5.00	4.00	B	
Q.10	A	B	C	D						
	P,Q	R,S	R,S	R,S						

EXERCISE # O-4

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	63	0	8	1	8	9.00	-10.00	3.00	1.00	-2.00

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	2	2	4	2	1	3	4	1	2
Que.	11	12	13							
Ans.	A	B	A							

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	2	C	B,C	B,D	A,B,C	B,C	D	D	D
Que.	11	12	13	14						
Ans.	C	5.00	C	A,B						

EXERCISE (O-1)

Straight Objective Type

1. The least area of a circle circumscribing any right triangle of area S is :

(A) πS (B) $2\pi S$ (C) $\sqrt{2} \pi S$ (D) $4 \pi S$

MM0001

2. A minimum value of $\sin x \cos 2x$ is-

(A) 1 (B) -1 (C) $-2/3 \sqrt{6}$ (D) None of these

MM0002

3. $f(x) = 1 + [\cos x]x$, in $0 \leq x \leq \frac{\pi}{2}$ (where $[.]$ denotes greatest integer function)

(A) has a minimum value 0 (B) has a maximum value 2
(C) is continuous in $\left[0, \frac{\pi}{2}\right]$ (D) is not differentiable at $x = \frac{\pi}{2}$

MM0003

4. Minimum value of the function $f(x) = \sum_{k=1}^5 (x-k)^2$ is at-

(A) $x = 2$ (B) $x = 5/2$ (C) $x = 3$ (D) $x = 5$

MM0004

5. If $f(x) = x^3 + ax^2 + bx + c$ is minimum at $x = 3$ and maximum at $x = -1$, then-

(A) $a = -3, b = -9, c = 0$ (B) $a = 3, b = 9, c = 0$
(C) $a = -3, b = -9, c \in \mathbb{R}$ (D) none of these

MM0005

6. The greatest value of $x^3 - 18x^2 + 96x$ in the interval $(0, 9)$ is-

(A) 128 (B) 60 (C) 160 (D) 120

MM0006

7. Difference between the greatest and the least values of the function $f(x) = x(\ln x - 2)$ on $[1, e^2]$ is

(A) 2 (B) e (C) e^2 (D) 1

MM0007

8. The maximum area of a right angled triangle with hypotenuse h is :-

(A) $\frac{h^2}{\sqrt{2}}$ (B) $\frac{h^2}{2}$ (C) $\frac{h^2}{4}$ (D) $\frac{h^2}{2\sqrt{2}}$

MM0008

9. If $f(x) = |x| + |x - 1| + |x - 2|$, then-
- (A) $f(x)$ has minima at $x = 1$ (B) $f(x)$ has maxima at $x = 0$
 (C) Equation $f(x) = 1$ has 1 solution (D) None of these

MM0009

10. Consider the function $f(x) = x \cos x - \sin x$, then identify the statement which is correct.
- (A) f is neither odd nor even
 (B) f is monotonic decreasing at $x = 0$
 (C) f has a maxima at $x = \pi$
 (D) f has a minima at $x = -\pi$

MM0010

11. Range of the function $f(x) = \frac{\ln x}{\sqrt{x}}$ is
- (A) $(-\infty, e)$ (B) $(-\infty, e^2)$ (C) $\left(-\infty, \frac{2}{e}\right]$ (D) $\left(-\infty, \frac{1}{e}\right)$

MM0011

12. If $(x - a)^{2m}(x - b)^{2n+1}$, where m and n are positive integers and $a > b$, is the derivative of a function f , then-
- (A) $x = a$ gives neither a maximum, nor a minimum
 (B) $x = a$ gives a maximum
 (C) $x = b$ gives neither a maximum nor a minimum
 (D) None of these

MM0012

13. The minimum value of $a \sec x + b \operatorname{cosec} x$, $0 < a < b$, $0 < x < \pi/2$ is-
- (A) $a + b$ (B) $a^{2/3} + b^{2/3}$ (C) $(a^{2/3} + b^{2/3})^{3/2}$ (D) None of these

MM0013

14. The rate of change of the function $f(x) = 3x^5 - 5x^3 + 5x - 7$ is minimum when
- (A) $x = 0$ (B) $x = 1/\sqrt{2}$ (C) $x = -1/\sqrt{2}$ (D) $x = \pm 1/\sqrt{2}$

MM0014

15. The cost of running a bus from A to B, is Rs. $\left(av + \frac{b}{v}\right)$, where v km/h is the average speed of the bus. When the bus travels at 30 km/h, the cost comes out to be Rs. 75 while at 40 km/h, it is Rs. 65. Then the most economical speed (in km/h) of the bus is :

- (A) 40 (B) 60 (C) 45 (D) 50

MM0015

16. The sum of lengths of the hypotenuse and another side of a right angled triangle is given. The area of the triangle will be maximum if the angle between them is :

(A) $\pi/6$ (B) $\pi/4$ (C) $\pi/3$ (D) $5\pi/12$

MM0016

17. If $ax + \frac{b}{x} \geq c$ for all positive x , where $a, b, c > 0$, then-

(A) $ab < \frac{c^2}{4}$ (B) $ab \geq \frac{c^2}{4}$ (C) $ab \geq \frac{c}{4}$ (D) None of these

MM0017

18. Two sides of a triangle are to have lengths 'a' cm & 'b' cm. If the triangle is to have the maximum area, then the length of the median from the vertex containing the sides 'a' and 'b' is

(A) $\frac{1}{2}\sqrt{a^2 + b^2}$ (B) $\frac{2a + b}{3}$ (C) $\sqrt{\frac{a^2 + b^2}{2}}$ (D) $\frac{a + 2b}{3}$

MM0018

19. A rectangle has one side on the positive y-axis and one side on the positive x - axis. The upper right hand vertex of the rectangle lies on the curve $y = \frac{\ln x}{x^2}$. The maximum area of the rectangle is

(A) e^{-1} (B) $e^{-1/2}$ (C) 1 (D) $e^{1/2}$

MM0019

20. P is a point on positive x-axis, Q is a point on the positive y-axis and 'O' is the origin. If the line passing through P and Q is tangent to the curve $y = 3 - x^2$ then the minimum area of the triangle OPQ, is

(A) 2 (B) 4 (C) 8 (D) 9

MM0020

21. The set of value(s) of 'a' for which the function $f(x) = \frac{ax^3}{3} + (a+2)x^2 + (a-1)x + 2$ posses a negative point of inflection.

(A) $(-\infty, -2) \cup (0, \infty)$ (B) $\{-4/5\}$
(C) $(-2, 0)$ (D) empty set

MM0021

22. If $f(x) = \begin{cases} e^{x+1} - e^x & x \leq 0 \\ e^{1-x} - 1 & 0 < x < 1, \text{ then -} \\ x + \ln x & x \geq 1 \end{cases}$

- (A) $x = 0$ is point of local maxima, $x = 1$ is neither local maxima nor local minima.
 (B) $x = 1$ is point of local minima, $x = 0$ is point of local maxima
 (C) $x = 0$ and $x = 1$ both are points of local maxima
 (D) $x = 0$ and $x = 1$ both are points of local minima

MM0022

23. Let $f(x) = ax^2 - b|x|$, where a and b are constants. Then at $x = 0$, $f(x)$ has

- (A) a local maxima whenever $a > 0$, $b > 0$
 (B) a local maxima whenever $a > 0$, $b < 0$
 (C) a local minima whenever $a > 0$, $b > 0$
 (D) neither a local maxima nor local minima whenever $a > 0$, $b < 0$

MM0023

24. The bottom of the legs of a three legged table are the vertices of an isosceles triangle with sides 5, 5 and 6. The legs are to be braced at the bottom by three wires in the shape of a Y. The minimum length of the wire needed for this purpose, is

- (A) $4 + 3\sqrt{3}$ (B) 10 (C) $3 + 4\sqrt{3}$ (D) $1 + 6\sqrt{2}$

MM0024

25. Let $f(x) = \frac{\tan^n x}{1 + \sum_{r=1}^{2n} \tan^r x}$, $n \in \mathbb{N}$, where $x \in \left[0, \frac{\pi}{2}\right)$

- (A) $f(x)$ is bounded and it takes both of it's bounds and the range of $f(x)$ contains exactly one integral point.
 (B) $f(x)$ is bounded and it takes both of it's bounds and the range of $f(x)$ contains more than one integral point.
 (C) $f(x)$ is bounded but minimum and maximum does not exists.
 (D) $f(x)$ is not bounded as the upper bound does not exist.

MM0025

26. If (a, b) be the point on the curve $y = |x^2 - 4x + 3|$ which is nearest to the circle $x^2 + y^2 - 4x - 4y + 7 = 0$, then $(a + b)$ is equal to -

- (A) $\frac{7}{4}$ (B) 0 (C) 3 (D) 2

MM0026

27. Let $f(x) = \begin{cases} x^{3/5} & \text{if } x \leq 1 \\ -(x-2)^3 & \text{if } x > 1 \end{cases}$, then the number of critical points on the graph of the function is
- (A) 1 (B) 2 (C) 3 (D) 4

MM0027

28. The set of all values of 'a' for which the function,

$$f(x) = (a^2 - 3a + 2) \left(\cos^2 \frac{x}{4} - \sin^2 \frac{x}{4} \right) + (a-1)x + \sin 1$$

does not possess critical points is :

- (A) $[1, \infty)$ (B) $(0, 1) \cup (1, 4)$ (C) $(-2, 4)$ (D) $(1, 3) \cup (3, 5)$

MM0028

29. There are 50 apple trees in an orchard. Each tree produces 800 apples. For each additional tree planted in the orchard, per tree the output drop by 10 times the number of additional trees. Number of trees that should be added to the existing orchard for maximising the output of the trees, is
- (A) 5 (B) 10 (C) 15 (D) 20

MM0029

30. Give the correct order of initials **T** or **F** for following statements. Use **T** if statement is true and **F** if it is false.

Statement-1: If $f: \mathbb{R} \rightarrow \mathbb{R}$ and $c \in \mathbb{R}$ is such that f is increasing in $(c - \delta, c)$ and f is decreasing in $(c, c + \delta)$ then f has a local maximum at c . Where δ is a sufficiently small positive quantity.

Statement-2: Let $f: (a, b) \rightarrow \mathbb{R}$, $c \in (a, b)$. Then f can not have both a local maximum and a point of inflection at $x = c$.

Statement-3: The function $f(x) = x^2 |x|$ is twice differentiable at $x = 0$.

Statement-4: Let $f: [c-1, c+1] \rightarrow [a, b]$ be bijective map such that f is differentiable at c then f^{-1} is also differentiable at $f(c)$.

- (A) FFTF (B) TTFT (C) FTTF (D) TTTF

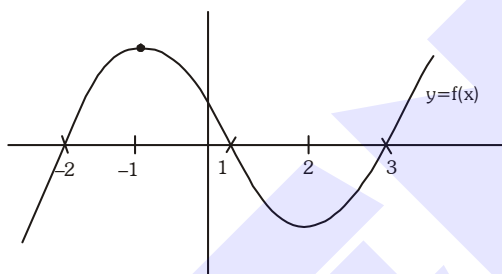
MM0030

EXERCISE (O-2)

Multiple Correct Answer Type

1. Let $f(x) = ax^3 + bx^2 + cx + 1$ have extrema at $x = \alpha, \beta$ such that $\alpha\beta < 0$ and $f(\alpha) \cdot f(\beta) < 0$, then which of the following can be true for the equation $f(x) = 0$?
- (A) three equal roots
- (B) three distinct real roots
- (C) one positive root if $f(\alpha) < 0$ and $f(\beta) > 0$
- (D) one negative root if $f(\alpha) > 0$ and $f(\beta) < 0$.
2. Graph of $y = f(x)$ is given, then
- MM0031**

MM0031



- (A) $y = H(x) = \max\{f(t) : t \leq x\} \forall x \in \mathbb{R}$ has a local maximum as well as a local minimum
- (B) $y = |f(x)|$ has 5 point of extrema
- (C) $y = f(|x|)$ has 5 points of extrema
- (D) $y = f(|x|)$ has 3 points of extrema

MM0032

3. If $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} [f(x)]$ ('a' is a finite quantity), where $[.]$ denotes greatest integer function and $f(x)$ is a non constant continuous function, then

- (A) $\lim_{x \rightarrow a} f(x)$ is an integer. (B) $\lim_{x \rightarrow a} f(x)$ need not be an integer.
- (C) $f(x)$ has a local minimum at $x = a$ (D) $f(x)$ has a local maximum at $x = a$.

MM0033

4. Let $f(x)$ be a cubic polynomial such that it has point of inflection at $x = 2$ and local minima at $x = 4$, then-
- (A) $f(x)$ has local minima at $x = 0$ (B) $f(x)$ has local maxima at $x = 0$
- (C) $\lim_{x \rightarrow \infty} f(x) \rightarrow \infty$ (D) $\lim_{x \rightarrow \infty} f(x) \rightarrow -\infty$

MM0034

5. Let $f(x, y) = \sqrt{x^2 + y^2} + \sqrt{x^2 + y^2 - 2x + 1} + \sqrt{x^2 + y^2 - 2y + 1} + \sqrt{x^2 + y^2 - 6x - 8y + 25} \quad \forall x, y \in \mathbb{R}$, then-

- (A) Minimum value of $f(x, y) = 5 + \sqrt{2}$ (B) Minimum value of $f(x, y) = 5 - \sqrt{2}$
 (C) Minimum value occurs of $f(x, y)$ for $x = \frac{3}{7}$ (D) Minimum value occurs of $f(x, y)$ for $y = \frac{4}{7}$

MM0035

6. Let $A(p, q)$ and $B(h, k)$ are points on the curve $4x^2 + 9y^2 = 1$, which are nearest and farthest from the line $72 + 8x = 9y$ respectively, then -

- (A) $p + q = -\frac{1}{5}$ (B) $p + q = \frac{1}{5}$ (C) $h + k = -\frac{1}{5}$ (D) $h + k = \frac{1}{5}$

MM0036

7. Let $f(x) = (x^2 - 1)^n (x^2 + x + 1)$. $f(x)$ has local extremum at $x = 1$ if

- (A) $n = 2$ (B) $n = 3$ (C) $n = 4$ (D) $n = 6$

MM0108

8. If $f(x) = \frac{x}{1 + x \tan x}$, $x \in \left(0, \frac{\pi}{2}\right)$, then

- (A) $f(x)$ has exactly one point of minima (B) $f(x)$ has exactly one point of maxima
 (C) $f(x)$ is increasing in $\left(0, \frac{\pi}{2}\right)$ (D) maxima occurs at x_0 where $x_0 = \cos x_0$

MM0109

9. If $f(x) = \begin{cases} -\sqrt{1-x^2} & , 0 \leq x \leq 1 \\ -x & , x > 1 \end{cases}$, then

- (A) Maximum of $f(x)$ exist at $x = 1$ (B) Maximum of $f(x)$ doesn't exist
 (C) Minimum of $f^{-1}(x)$ exist at $x = -1$ (D) Minimum of $f^{-1}(x)$ exist at $x = 1$

MM0110

10. If the function $y = f(x)$ is represented as,

$$x = \phi(t) = t^3 - 5t^2 - 20t + 7$$

$$y = \psi(t) = 4t^3 - 3t^2 - 18t + 3 \quad (|t| < 2), \text{ then :}$$

- (A) $y_{\max} = 12$ (B) $y_{\max} = 14$ (C) $y_{\min} = -67/4$ (D) $y_{\min} = -69/4$

MM0111

11. If $f(x) = \tan^{-1}x - (1/2) \ln x$. Then

- (A) the greatest value of $f(x)$ on $\left[1/\sqrt{3}, \sqrt{3}\right]$ is $\pi/6 + (1/4) \ln 3$
 (B) the least value of $f(x)$ on $\left[1/\sqrt{3}, \sqrt{3}\right]$ is $\pi/3 - (1/4) \ln 3$
 (C) $f(x)$ decreases on $(0, \infty)$
 (D) $f(x)$ increases on $(-\infty, 0)$

MM0112

12. If P is a point on the curve $5x^2 + 3xy + y^2 = 2$ and O is the origin, then OP has

- (A) minimum value $\frac{1}{2}$ (B) minimum value $\frac{2}{\sqrt{11}}$
 (C) maximum value $\sqrt{11}$ (D) maximum value 2

MM0113

13. Let $f(x) = \sqrt{(1-x^4)(1+4x^2)}$, then choose the correct option(s) -

- (A) the greatest value of $f(x)$ is 1 (B) the greatest value of $f(x)$ is $\frac{3}{2}$
 (C) the least value of $f(x)$ is 0 (D) the least value of $f(x)$ is 1

MM0114

14. Let $f(x) = \min\{|x|, \sqrt{1-x^2}\}; x \in [-1, 1]$ and $g(x) = \max\{|x|, \sqrt{1-x^2}\}; x \in [-1, 1]$. Which of the following statement(s) is (are) correct?

- (A) number of local minima of $f(x)$ is 1 (B) number of local minima of $f(x)$ is 3
 (C) $x = 0$ is a point of local maxima for $g(x)$ (D) number of local maxima of $g(x)$ is 3

MM0115

15. Let $f(x) = x \sin x - \cos x$ be defined over the set A, where $A = \left\{x : x \log\left(\frac{2x}{\pi}\right) \leq 0\right\}$, then

- (A) maximum value of $f(x)$ is $\frac{\pi}{2}$ (B) minimum value of $f(x)$ is -1
 (C) minimum value of $f(x)$ is -2 (D) number of solution of $|f(x)| = 0.5$ is 2

MM0116

EXERCISE (O-3)**Linked Comprehension Type****Paragraph for Question 1 to 3**

Let $f(x) = 1 + a^2x - x^3$ where 'a' is real number, point of minima of $f(x)$ must lie between $[A, B]$

where A and B is the minimum and maximum value of $\frac{x^2 + 3x + 1}{x^2 + x + 1}$ for all $x \in \mathbb{R}$.

On the basis of above information, answer the following questions-

1. Find the value of A

(A) -1 (B) $-\frac{1}{2}$ (C) -2 (D) None of these

MM0117

2. Find the value of B

(A) $\frac{3}{2}$ (B) $\frac{5}{3}$ (C) $\frac{5}{4}$ (D) None of these

MM0118

3. If $a > 0$, then find the point of local minima

(A) $\frac{-a}{\sqrt{3}}$ (B) $\frac{a}{\sqrt{3}}$ (C) $\frac{a}{\sqrt{2}}$ (D) None of these

MM0119

Paragraph for Question 4 and 5

Let $f : \mathbb{R} \rightarrow \mathbb{R}$, $g : \mathbb{R} \rightarrow \mathbb{R}$, $h : \mathbb{R} \rightarrow \mathbb{R}$

$f(x) = x^2e^x$, $g(x) = x^3 - x^2 + x$ and $h(x) = x + 2$.

4. Choose the correct statement (s)

(A) The local minimum value of $f(g(x))$ is 0
 (B) The local minimum value of $f(g(x))$ is an odd number
 (C) The local minimum value of $f(g(x))$ does not exist
 (D) The local minimum value of $f(g(x))$ is an even number

MM0120

5. Choose the correct statement (s)

(A) The local minimum value of $h(f(x))$ is 0
 (B) The local minimum value of $h(f(x))$ is an odd number
 (C) The local minimum value of $h(f(x))$ is 2
 (D) The local minimum value of $h(f(x))$ is an even number

MM0121

Paragraph for Question 6 to 8

Consider $f(x) = x^3(x - 2)^2(x - 1)$.

6. If least value of $f(x)$ is at $x = a$, then the value of $(3a - 2)$ is MM0122
7. If least value of $f(x)$ is L , then the value of $[L]$ is MM0123
(where $[.]$ represents greatest integer function)
8. If maxima and minima of $f(x)$ occurs at $x_i, i \in N$, then $\left[\sum_{i=1}^n x_i^2 \right]$ is MM0124
(where $[.]$ represents greatest integer function)

Matrix Match Type

9. **Column-I** **Column-II**
- (A) The number of point (s) of maxima of $f(x) = x^2 + \frac{1}{x^2}$ is (P) 0
- (B) $(\sin^{-1}x)^3 + (\cos^{-1}x)^3$ is maximum at $x =$ (Q) 2
- (C) If $[a, b]$, $(b < 1)$ is largest interval in which (R) $\frac{8}{3}$
 $f(x) = 3x^4 + 8x^3 - 6x^2 - 24x + 19$ is strictly
increasing then $\frac{a}{b}$ is
- (D) If $a + b = 8$, $a, b > 0$ then minimum value of $\frac{a^3 + b^3}{48}$ is (S) -1 MM0125
10. **Column-I** **Column-II**
- (A) A rectangle is inscribed in an equilateral triangle of side 4cm. Square of maximum area of such a rectangle is (P) 65
- (B) The volume of a rectangular closed box is 72 and the base sides are in the ratio 1 : 2. The least total surface area is (Q) 45
- (C) If x and y are two positive numbers such that $x + y = 60$ and x^3y is maximum then value of x is (R) 12
- (D) The sides of a rectangle of greatest perimeter which is inscribed in a semicircle of radius $\sqrt{5}$ are a and b . Then $a^3 + b^3 =$ (S) 108 MM0126

EXERCISE (O-4)

Numerical Grid Type

1. The graph of the derivative $f'(x)$ of a continuous function $f(x)$ in $(0, 9)$. If

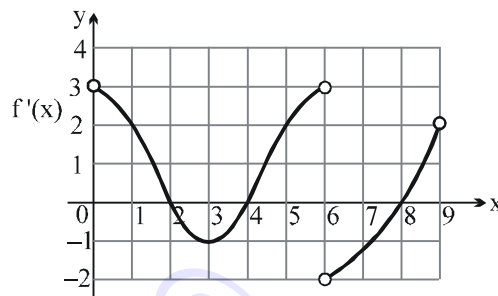
(i) f is strictly increasing in the interval $(a, b]$; $[c, d]$; $[e, f)$ and strictly decreasing in $[p, q]$; $[r, s]$.

(ii) f has a local minima at $x = x_1$ and $x = x_2$.

(iii) $f''(x) > 0$ in (l, m) ; (n, t)

(iv) f has inflection point at $x = k$

(v) number of critical points of $y = f(x)$ is 'w'.



Find the value of $(a + b + c + d + e) + (p + q + r + s) + (l + m + n) + (x_1 + x_2) + (k + w)$.

MM0037

2. Let $f(x)$ be a cubic polynomial which has local maximum at $x = -1$ and $f'(x)$ has a local minimum at $x = 1$. If $f(-1) = 10$ and $f(3) = -22$, then find the distance between its two horizontal tangents.

MM0038

3. Let $P(x)$ be a polynomial of degree 5 having extremum at $x = -1, 1$ and $\lim_{x \rightarrow 0} \left(\frac{P(x)}{x^3} - 2 \right) = 4$.

If M and m are the maximum and minimum value of the function $y = P'(x)$ on the set

$A = \{x | x^2 + 6 \leq 5x\}$ then find $\frac{m}{M}$.

MM0042

4. Find the area of the largest rectangle with lower base on the x -axis & upper vertices on the curve $y = 12 - x^2$.

MM0044

5. Of all the lines tangent to the graph of the curve $y = \frac{6}{x^2 + 3}$, then the sum of the minimum and maximum slopes of these tangent lines.

MM0045

6. The length of three sides of a trapezium are equal, each being 10 cms. If the maximum area of such a trapezium is A sq. units, then the value of $(\sqrt{3} A)$ is.

MM0047

7. By the post office regulations, the combined length & girth of a parcel must not exceed 3 metre. If the volume of the biggest cylindrical (right circular) packet that can be sent by the parcel post is V cu.m, then the value of $(4\pi V)$ is

MM0048

8. A running track of 440 ft. is to be laid out enclosing a football field, the shape of which is a rectangle with semi circle at each end. If the area of the rectangular portion is to be maximum, then the perimeter of the rectangular portion is

Use : $\pi \approx 22/7$.

MM0049

9. The value of 'a' for which $f(x) = x^3 + 3(a-7)x^2 + 3(a^2-9)x - 1$ have a positive point of maximum lies in the interval $(a_1, a_2) \cup (a_3, a_4)$. Find the value of $a_2 + 11a_3 + 70a_4$.

MM0053

10. Given two points A $(-2, 0)$ & B $(0, 4)$ and a line $y = x$. If the co-ordinates of a point M on this line is (p, q) so that the perimeter of the ΔAMB is least, then the value of $(5p + 3q)$ is

MM0058

EXERCISE (JM)

1. Twenty meters of wire is available for fencing off a flower-bed in the form of a circular sector. Then the maximum area (in sq. m) of the flower -bed, is :- [JEE-MAIN 2017]

(1) 30 (2) 12.5 (3) 10 (4) 25

MM0076

2. Let $f(x) = x^2 + \frac{1}{x^2}$ and $g(x) = x - \frac{1}{x}$, $x \in \mathbb{R} - \{-1, 0, 1\}$. If $h(x) = \frac{f(x)}{g(x)}$, then the local minimum value of $h(x)$ is : [JEE-MAIN 2018]

(1) -3 (2) $-2\sqrt{2}$ (3) $2\sqrt{2}$ (4) 3

MM0077

3. A helicopter is flying along the curve given by $y - x^{3/2} = 7$, ($x \geq 0$). A soldier positioned at the point $\left(\frac{1}{2}, 7\right)$ wants to shoot down the helicopter when it is nearest to him. Then this nearest distance is : [JEE-MAIN 2019]

(1) $\frac{1}{2}$ (2) $\frac{1}{3}\sqrt{\frac{7}{3}}$ (3) $\frac{1}{6}\sqrt{\frac{7}{3}}$ (4) $\frac{\sqrt{5}}{6}$

MM0078

4. Let $a_1, a_2, a_3, \dots$ be an A. P. with $a_6 = 2$. Then the common difference of this A. P., which maximises the produce $a_1 a_4 a_5$, is : [JEE-MAIN 2019]

(1) $\frac{6}{5}$ (2) $\frac{8}{5}$ (3) $\frac{2}{3}$ (4) $\frac{3}{2}$

MM0079

5. Let $f(x)$ be a polynomial of degree 5 such that $x = \pm 1$ are its critical points. If $\lim_{x \rightarrow 0} \left(2 + \frac{f(x)}{x^3}\right) = 4$, then which one of the following is not true ? [JEE-MAIN 2020]

(1) f is an odd function
 (2) $x = 1$ is a point of minima and $x = -1$ is a point of maxima of f .
 (3) $x = 1$ is a point of maxima and $x = -1$ is a point of minimum of f .
 (4) $f(1) - 4f(-1) = 4$

MM0080

6. Let $f(x)$ be a polynomial of degree 3 such that $f(-1) = 10$, $f(1) = -6$, $f(x)$ has a critical point at $x = -1$ and $f'(x)$ has a critical point at $x = 1$. Then $f(x)$ has a local minima at $x =$ _____. [JEE-MAIN 2020]

MM0081

7. A spherical iron ball of 10 cm radius is coated with a layer of ice of uniform thickness the melts at a rate of $50 \text{ cm}^3/\text{min}$. When the thickness of ice is 5 cm, then the rate (in cm/min .) at which of the thickness of ice decreases, is : [JEE-MAIN 2020]

- (1) $\frac{1}{36\pi}$ (2) $\frac{5}{6\pi}$ (3) $\frac{1}{18\pi}$ (4) $\frac{1}{54\pi}$

MM0082

8. The triangle of maximum area that can be inscribed in a given circle of radius 'r' is :

(1) An isosceles triangle with base equal to $2r$. [JEE-MAIN 2021]

(2) An equilateral triangle of height $\frac{2r}{3}$.

(3) An equilateral triangle having each of its side of length $\sqrt{3}r$.

(4) A right angle triangle having two of its sides of length $2r$ and r . MM0127

9. The sum of all the local minimum values of the twice differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined

by $f(x) = x^3 - 3x^2 - \frac{3f''(2)}{2}x + f''(1)$ is :

[JEE-MAIN 2021]

- (1) -22 (2) 5 (3) -27 (4) 0

MM0128

10. The sum of the absolute minimum and the absolute maximum values of the function $f(x) = |3x - x^2 + 2| - x$ in the interval $[-1, 2]$ is : [JEE-MAIN 2021]

- (A) $\frac{\sqrt{17}+3}{2}$ (B) $\frac{\sqrt{17}+5}{2}$ (C) 5 (D) $\frac{9-\sqrt{17}}{2}$

MM0129

11. The curve $y(x) = ax^3 + bx^2 + cx + 5$ touches the x-axis at the point $P(-2, 0)$ and cuts the y-axis at the point Q, where y' is equal to 3. Then the local maximum value of $y(x)$ is :

- (A) $\frac{27}{4}$ (B) $\frac{29}{4}$ (C) $\frac{37}{4}$ (D) $\frac{9}{2}$

[JEE-MAIN 2022]

MM0130

12. A water tank has the shape of a right circular cone with axis vertical and vertex downwards. Its semi-vertical angle is $\tan^{-1} \frac{3}{4}$. Water is poured in it at a constant rate of 6 cubic meter per hour.

The rate (in square meter per hour), at which the wet curved surface area of the tank is increasing, when the depth of water in the tank is 4 meters, is _____.

[JEE-MAIN 2022]

MM0131

13. Let $x = 2$ be a local minima of the function $f(x) = 2x^4 - 18x^2 + 8x + 12$, $x \in (-4, 4)$. If M is local maximum value of the function f in $(-4, 4)$, then M =

- (1) $12\sqrt{6} - \frac{33}{2}$ (2) $12\sqrt{6} - \frac{31}{2}$ (3) $18\sqrt{6} - \frac{33}{2}$ (4) $18\sqrt{6} - \frac{31}{2}$

[JEE-MAIN 2023]

MM0132

EXERCISE (JA)

1. (a) Let f , g and h be real-valued functions defined on the interval $[0, 1]$ by $f(x) = e^{x^2} + e^{-x^2}$, $g(x) = xe^{x^2} + e^{-x^2}$ and $h(x) = x^2e^{x^2} + e^{-x^2}$. If a , b and c denote respectively, the absolute maximum of f , g and h on $[0, 1]$, then
 (A) $a = b$ and $c \neq b$ (B) $a = c$ and $a \neq b$ (C) $a \neq b$ and $c \neq b$ (D) $a = b = c$

MM0083

- (b) Let f be a function defined on $\mathbf{R}$ (the set of all real numbers) such that $f'(x) = 2010(x-2009)(x-2010)^2(x-2011)^3(x-2012)^4$, for all $x \in \mathbf{R}$.
 If g is a function defined on $\mathbf{R}$ with values in the interval $(0, \infty)$ such that $f(x) = \ln(g(x))$, for all $x \in \mathbf{R}$,
 then the number of points in $\mathbf{R}$ at which g has a local maximum is [JEE 2010, 3 + 3]

MM0084

2. Let $f: \mathbf{R} \rightarrow \mathbf{R}$ be defined as $f(x) = |x| + |x^2 - 1|$. The total number of points at which f attains either a local maximum or a local minimum is [JEE 2012, 4M]

MM0085

3. Let $p(x)$ be a real polynomial of least degree which has a local maximum at $x = 1$ and a local minimum at $x = 3$. If $p(1) = 6$ and $p(3) = 2$, then $p'(0)$ is [JEE 2012, 4M]

MM0086

4. A rectangular sheet of fixed perimeter with sides having their lengths in the ratio of 8 : 15 is converted into an open rectangular box by folding after removing squares of equal area from all four corners. If the total area of removed squares is 100, the resulting box has maximum volume. Then the lengths of the sides of the rectangular sheet are [JEE 2013, 4M, -1M]

(A) 24 (B) 32 (C) 45 (D) 60 MM0087

5. The function $f(x) = 2|x| + |x + 2| - ||x + 2| - 2|x||$ has a local minimum or a local maximum at $x =$ [JEE 2013, 3M, -1M]

(A) -2 (B) $-\frac{2}{3}$ (C) 2 (D) $\frac{2}{3}$

MM0088

6. A cylindrical container is to be made from certain solid material with the following constraints. It has a fixed inner volume of $V \text{ mm}^3$, has a 2 mm thick solid wall and is open at the top. The bottom of the container is a solid circular disc of thickness 2 mm and is of radius equal to the outer radius of the container.

If the volume of the material used to make the container is minimum when the inner radius of the

container is 10mm, then the value of $\frac{V}{250\pi}$ is [JEE 2015, 4M, 0M]

MM0089

7. The least value of $\alpha \in \mathbb{R}$ for which $4\alpha x^2 + \frac{1}{x} \geq 1$, for all $x > 0$, is -

[JEE(Advanced)-2016, 3(-1)]

- (A) $\frac{1}{64}$ (B) $\frac{1}{32}$ (C) $\frac{1}{27}$ (D) $\frac{1}{25}$

MM0090

8. Let $f : \mathbb{R} \rightarrow (0, \infty)$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ be twice differentiable function such that f'' and g'' are continuous functions on $\mathbb{R}$. Suppose $f'(2) = g(2) = 0$, $f''(2) \neq 0$ and $g'(2) \neq 0$. If $\lim_{x \rightarrow 2} \frac{f(x)g(x)}{f'(x)g'(x)} = 1$, then

[JEE(Advanced)-2016, 4(-2)]

- (A) f has a local minimum at $x = 2$
 (B) f has a local maximum at $x = 2$
 (C) $f''(2) > f(2)$
 (D) $f(x) - f''(x) = 0$ for at least one $x \in \mathbb{R}$

MM0091

9. If $f(x) = \begin{vmatrix} \cos(2x) & \cos(2x) & \sin(2x) \\ -\cos x & \cos x & -\sin x \\ \sin x & \sin x & \cos x \end{vmatrix}$, then

[JEE(Advanced)-2017]

- (A) $f'(x) = 0$ at exactly three points in $(-\pi, \pi)$
 (B) $f(x)$ attains its maximum at $x = 0$
 (C) $f(x)$ attains its minimum at $x = 0$
 (D) $f'(x) = 0$ at more than three points in $(-\pi, \pi)$

MM0092

10. For every twice differentiable function $f : \mathbb{R} \rightarrow [-2, 2]$ with $(f(0))^2 + (f'(0))^2 = 85$, which of the following statement(s) is (are) TRUE ?

[JEE(Advanced)-2018]

- (A) There exist $r, s \in \mathbb{R}$, where $r < s$, such that f is one-one on the open interval (r, s) .
 (B) There exists $x_0 \in (-4, 0)$ such that $|f'(x_0)| \leq 1$
 (C) $\lim_{x \rightarrow \infty} f(x) = 1$
 (D) There exists $\alpha \in (-4, 4)$ such that $f(\alpha) + f''(\alpha) = 0$ and $f'(\alpha) \neq 0$

MM0093

11. Let $f(x) = \frac{\sin \pi x}{x^2}$, $x > 0$

Let $x_1 < x_2 < x_3 < \dots < x_n < \dots$ be all the points of local maximum of f
and $y_1 < y_2 < y_3 < \dots < y_n < \dots$ be all the points of local minimum of f .

Then which of the following options is/are correct ?

[JEE(Advanced)-2019, 4(-1)]

- (1) $|x_n - y_n| > 1$ for every n (2) $x_1 < y_1$
(3) $x_n \in \left(2n, 2n + \frac{1}{2}\right)$ for every n (4) $x_{n+1} - x_n > 2$ for every n

MM0094

12. Consider all rectangles lying in the region

$$\left\{ (x, y) \in \mathbb{R} \times \mathbb{R} : 0 \leq x \leq \frac{\pi}{2} \text{ and } 0 \leq y \leq 2 \sin(2x) \right\}$$

and having one side on the x -axis. The area of the rectangle which has the maximum perimeter among all such rectangles, is

[JEE(Advanced)-2020]

- (A) $\frac{3\pi}{2}$ (B) π (C) $\frac{\pi}{2\sqrt{3}}$ (D) $\frac{\pi\sqrt{3}}{2}$

MM0105

13. Let the function $f: (0, \pi) \rightarrow \mathbb{R}$ be defined by

$$f(\theta) = (\sin \theta + \cos \theta)^2 + (\sin \theta - \cos \theta)^4$$

Suppose the function f has a local minimum at θ precisely when $\theta \in \{\lambda_1 \pi, \dots, \lambda_r \pi\}$,

where $0 < \lambda_1 < \dots < \lambda_r < 1$. Then the value of $\lambda_1 + \dots + \lambda_r$ is _____

[JEE(Advanced)-2020]

MM0106

Question Stem for Questions Nos. 14 and 15

Question Stem

Let $f_1: (0, \infty) \rightarrow \mathbb{R}$ and $f_2: (0, \infty) \rightarrow \mathbb{R}$ be defined by $f_1(x) = \int_0^x \prod_{j=1}^{21} (t-j)^j dt$, $x > 0$

and

$$f_2(x) = 98(x-1)^{50} - 600(x-1)^{49} + 2450, \quad x > 0,$$

where, for any positive integer n and real numbers $a_1, a_2, \dots, a_n$, $\prod_{i=1}^n a_i$ denotes the product of $a_1, a_2, \dots, a_n$. Let m_i and n_i , respectively, denote the number of points of local minima and the number of points of local maxima of function f_i , $i = 1, 2$, in the interval $(0, \infty)$ [JEE(Advanced)-2021]

14. The value of $2m_1 + 3n_1 + m_1n_1$ is _____.

MM0107

15. The value of $6m_2 + 4n_2 + 8m_2n_2$ is _____.

MM0108

16. Let $\alpha = \sum_{k=1}^{\infty} \sin^{2k} \left(\frac{\pi}{6} \right)$.

Let $g : [0, 1] \rightarrow \mathbb{R}$ be the function defined by

$$g(x) = 2^{\alpha x} + 2^{\alpha(1-x)}$$

Then, which of the following statements is/are TRUE ?

[JEE(Advanced)-2022]

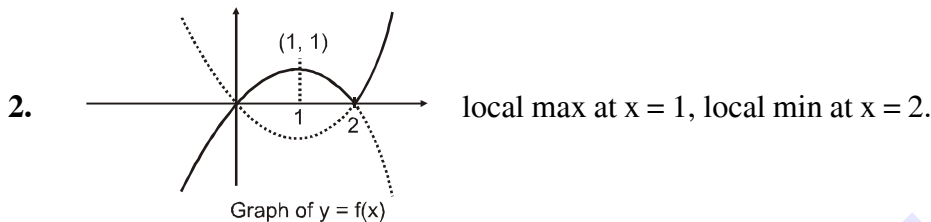
- (A) The minimum value of $g(x)$ is $2^{\frac{7}{6}}$
- (B) The maximum value of $g(x)$ is $1 + 2^{\frac{1}{3}}$
- (C) The function $g(x)$ attains its maximum at more than one point
- (D) The function $g(x)$ attains its minimum at more than one point

MM0133

ANSWER KEY

Do yourself - 1

1. (i) points of minima : $x = 0, x = 2$
 (ii) points of minima : $x \in (-\infty, 0]$; points of maxima : $x \in [2, \infty)$
 (iii) points of minima : $x = 0$



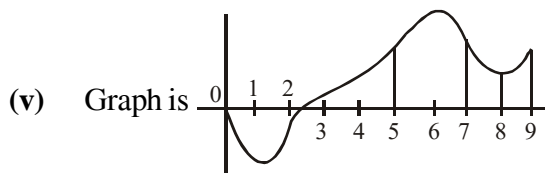
3. 3

Do yourself - 2

1. local max. at $x = -1$, local min. at $x = 1$
2. 121
3. (a) local max. at $x = -\frac{1}{5}$, local min. at $x = -1$
 (b) local mini at $x = \frac{1}{e}$, No local maxima
4. Local Minima at $x = \frac{\pi}{2} + 2n\pi, n \in \mathbb{I}$, Minima of $f(x) = 1$, Local Maxima at $x = -\frac{\pi}{2} + 2n\pi, n \in \mathbb{I}$,
 Maxima of $f(x) = -1$
5. $b \in (0, e]$
6. 4
7. 3
8. 3
9. $a = 1, b = 0$
10. $x = \frac{\pi}{4}$

Do yourself - 3

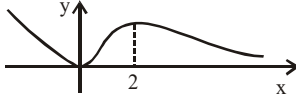
1. $1/e$
2. $1/e^2$
3. local minima at $x = 3$
4. (i) I in $[1, 6]$; $[8, 9]$ and D in $[0, 1]$; $[6, 8]$;
 (ii) L.Min. at $x = 1$ and $x = 8$; L.Max. at $x = 0, 6, 9$
 (iii) $(0, 2) \cup (3, 5) \cup (7, 9)$ and $(2, 3) \cup (5, 7)$;
 (iv) $x = 2, 3, 5, 7$



5. (a) strictly increasing in $[0, 2]$ and strictly decreasing in $(-\infty, 0]$; $[2, \infty)$, local min. value = 0 and local max. value = 2

- (b) concave up for $(-\infty, 2 - \sqrt{2}]$; $[2 + \sqrt{2}, \infty)$ and concave down in $[2 - \sqrt{2}, 2 + \sqrt{2}]$

(c) $f(x) = \frac{1}{2}e^{2-x} \cdot x^2$



Do yourself - 4

1. $x = 15$ & $y = 45$
2. 9 & 9
3. 24
4. $r = \left(\frac{50}{\pi}\right)^{1/3}$ cm. & $h = 2\left(\frac{50}{\pi}\right)^{1/3}$ cm.
5. 4
6. 4
7. $r = \sqrt{\frac{2A}{\pi+4}}$, $s = \sqrt{\frac{2A}{\pi+4}}$
8. side 10', height 10'
9. $P_{\max} = a\left(1 + \operatorname{cosec} \frac{\alpha}{2}\right)$
10. $\pi/4$

Do yourself - 5

1. (2, -4)
2. 4
3. $|a - b|$

Do yourself - 6

1. $x = 1$ is a critical point as well as stationary point (Note $x = 0$ is not in the domain of $f(x)$)
2. $x = 2$
3. (a) $(-\infty, 2) \cup (3, \infty)$ (b) (2, 3) (c) $x = 2$ & $x = 3$

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	B	C	C	C	C	B	C	A	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	A	C	D	B	C	B	A	A	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	A	A	A	A	A	C	C	B	C	C

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B,C,D	A,B,D	A,C	B,C	A,C,D	A,D	A,C,D	B,D	A,C	B,D
Que.	11	12	13	14	15					
Ans.	A,B,C	B,D	B,C	B,C,D	A,D					

EXERCISE # O-3

Que.	1	2	3	4	5	6	7	8		
Ans.	A	B	A	A,D	C,D	0.00	-1.00	6.00		
Q.9	A	B	C	D	Q.10	A	B	C	D	
	P	S	Q	R		R	S	Q	P	

EXERCISE # O-4

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	74	32	6	32	0	225.00	4.00	360.00	320.00	0.00

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	3	3	2	2	3	3	3	3	A
Que.	11	12	13							
Ans.	A	5	1							

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	(a) D, (b) 1	5	9	A,C	A,B	4	C	A,D	B,D	A,B,D
Que.	11	12	13	14	15	16				
Ans.	1,3,4	C	0.50	57.00	6.00	A,B,C				

Important Notes

